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MODULE INTRODUCTION



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Module Content



SEMESTER LONG
PROJECT



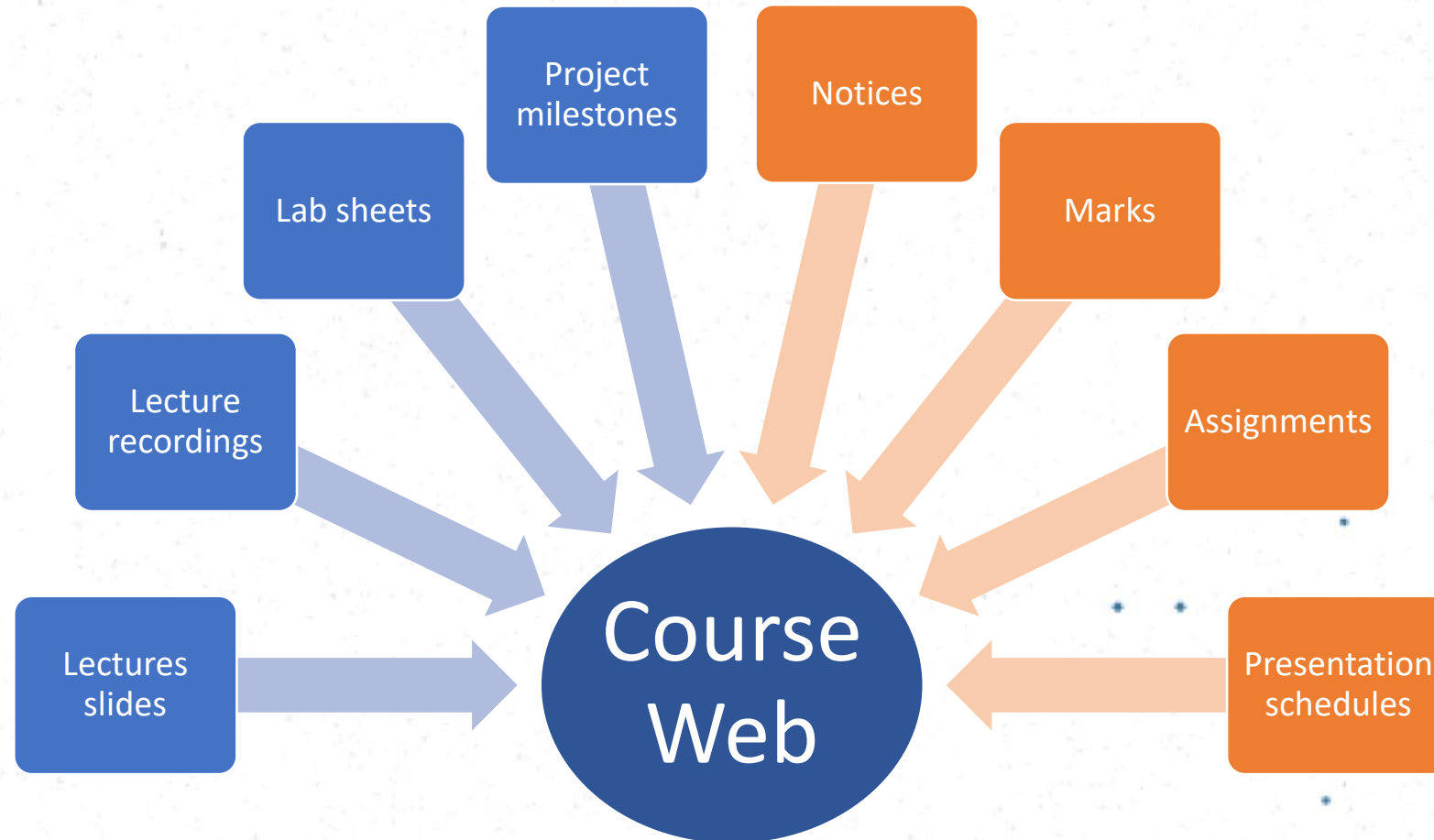
PROJECT MANAGEMENT
KNOWLEDGE AREAS



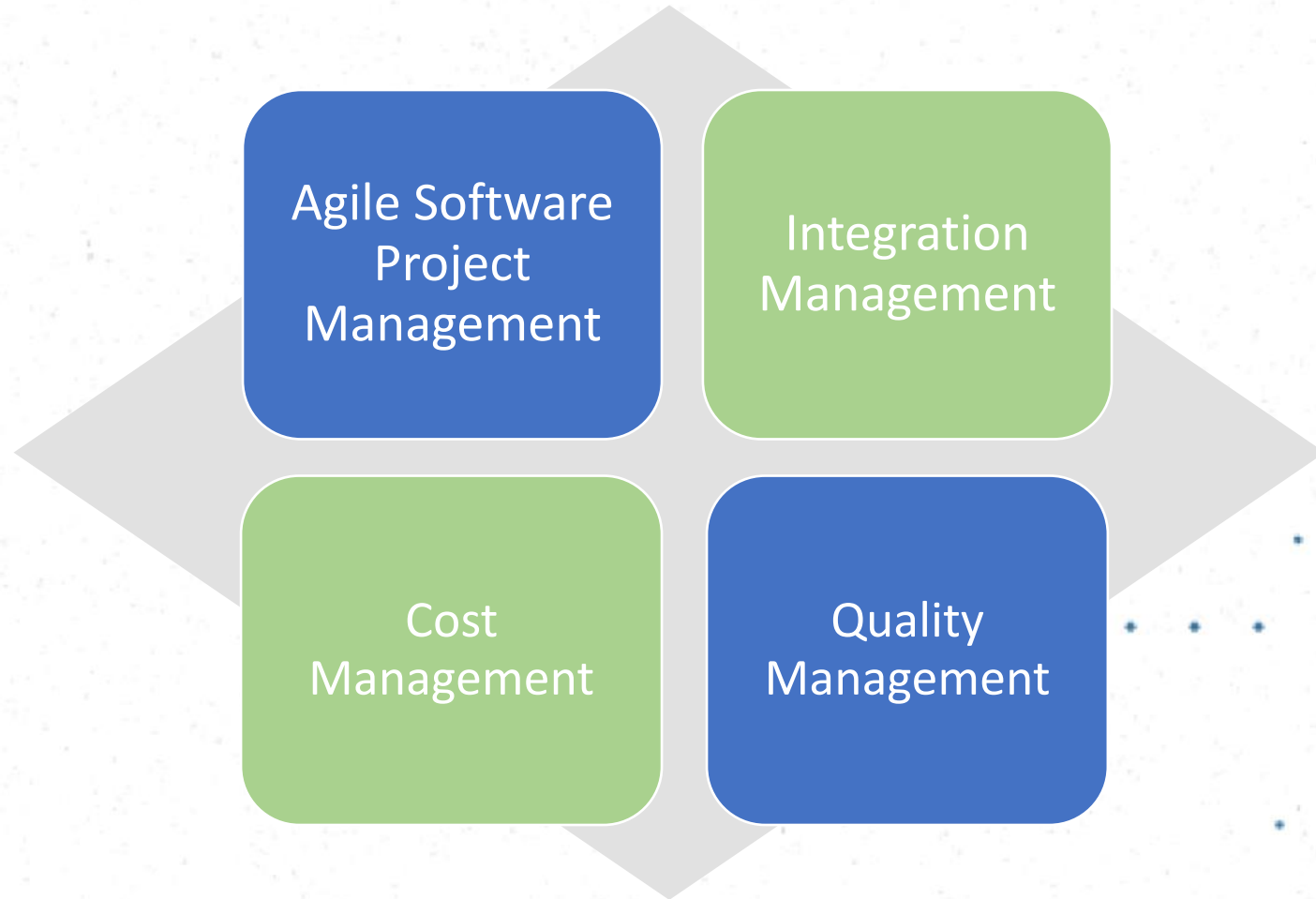
TOOLS FOR MANAGING
THE PROJECT

- 
- Team-working skills
 - Organizing and planning skills
 - Decision making skills
 - Time management skills
 - Leadership skills
- 
- Initiative skills
 - Analytical skills
 - Learning skills
 - Communication skills
 - Working under competitive conditions
- 

Resources and Materials



Lectures

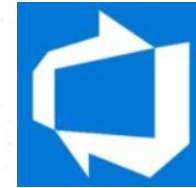


Lectures



Practicals

- Self-learning practicals on:
 - Azure Devops - Agile project management tool
 - SonarQube – Code quality inspection tool
 - Git – Distributed version control system
 - Selenium – Test automation frameworks





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Agile Software Project Management



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Who is a Project Manager?

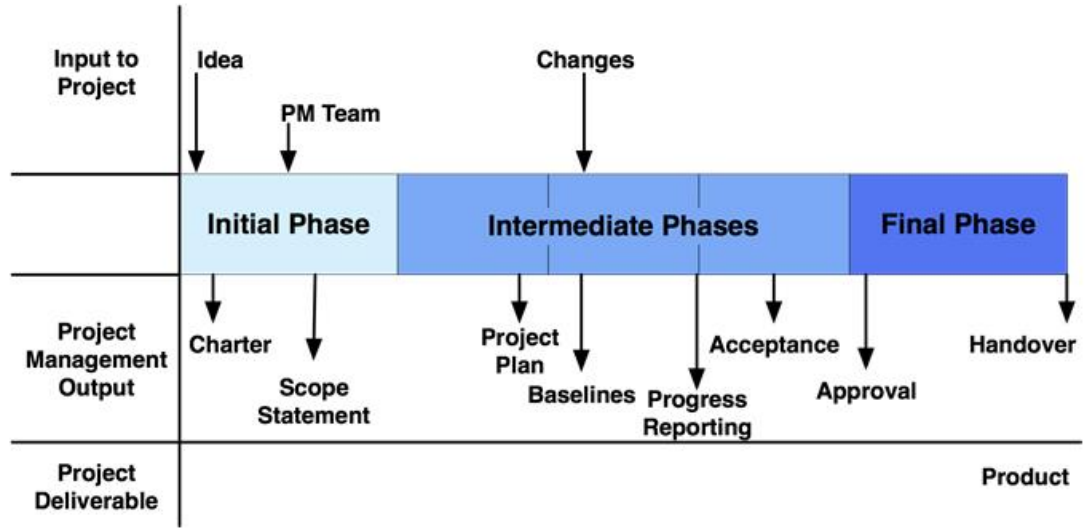
- Planning
- Scheduling
- Coordinating
- Working with people

Phases In a Project Life Cycle

The completion & approval of one or more deliverables defines a project phase

In **iterative systems development**, new phase can start without closing the previous phase.

A phase can be closed without initiating subsequent phases.



Adapted from PMBOK Guide 3rd Edition Fig. 2-3

Organizational Structures

Functional

Engineering, Marketing, Design, etc
P&L from production

Projectized

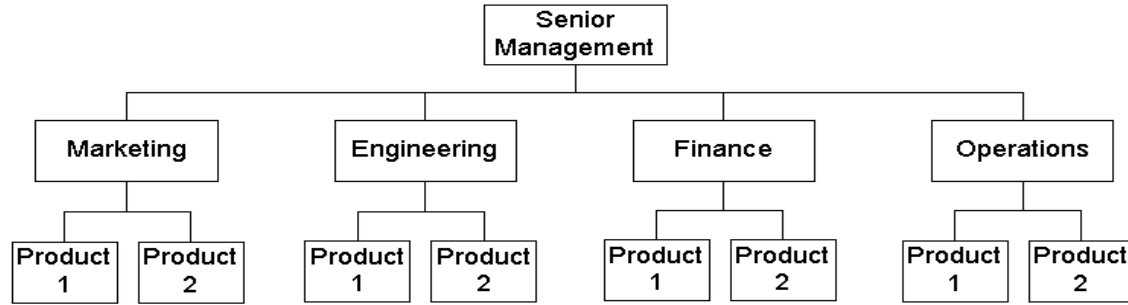
Project A, Project B
Income from projects
PM has P&L responsibility

Matrix

Functional and Project based
Program Mgmt. Model
Shorter cycles, need for rapid development process



Functional Organization



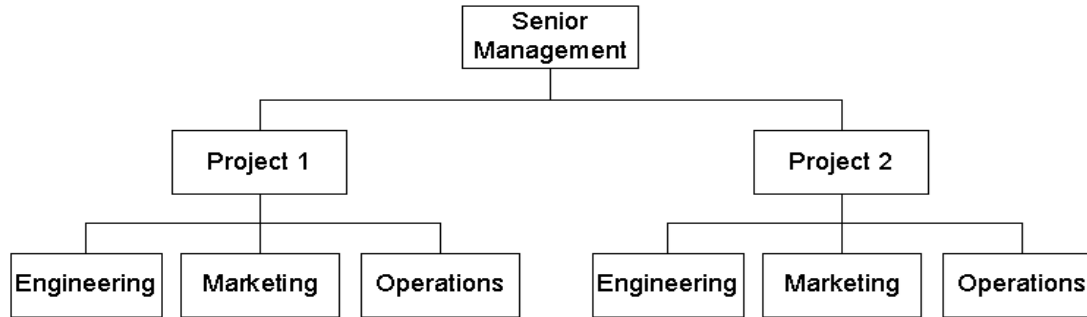
Pros

- Clear definition of authority
- Eliminates duplication
- Encourages specialization
- Clear career paths

Cons

- “Walls”: can lack customer orientation
- “Silos” create longer decisions cycles
- Conflicts across functional areas
- Project leaders have little power

Projectized Organization



Pros

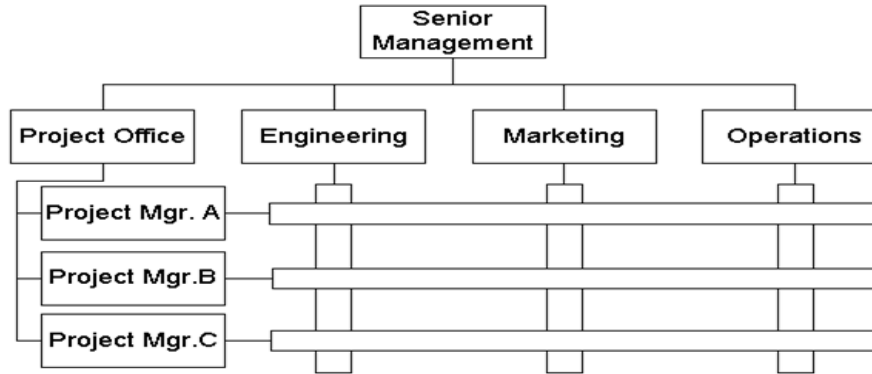
- Unity of command
- Effective intra-project communication

Cons

- Duplication of facilities
- Career path

Examples: defense avionics, construction

Matrix Organization



Pros

- Project integration across functional lines
- Efficient use of resources
- Retains functional teams

Cons

- Two bosses for personnel
- Complexity
- Resource & priority conflicts

Matrix Forms

Weak, Strong, Balanced **Degree of relative power**

Weak: functional-centric

Strong: project-centric

Organizational Structure – Influences on Projects

Organization Type Project Characteristics	Functional	Matrix			Projectized
		Weak Matrix	Balanced Matrix	Strong Matrix	
Project Manager's Authority	Little or None	Limited	Low to Moderate	Moderate To High	High to Almost Total
Percent of Performing Organization's Personnel Assigned Full-time to Project Work Project Manager's Role	Virtually None	0-25%	15-60%	50-95%	85-100%
	Part-time	Part-time	Full-time	Full-time	Full-time
Common Title for Project Manager's Role	Project Coordinator/ Project Leader	Project Coordinator/ Project Leader	Project Manager/ Project Officer	Project Manager/ Program Manager	Project Manager/ Program Manager
Project Management Administrative Staff	Part-time	Part-time	Part-time	Full-time	Full-time

Agile Transformation



<http://www.michele-joy.com/blog/2018/7/11/taking-the-leap>

Agile Manifesto

Individuals & Interactions
Over
Tools & Processes

Agile Manifesto

Working Software
Over
Comprehensive Documentation

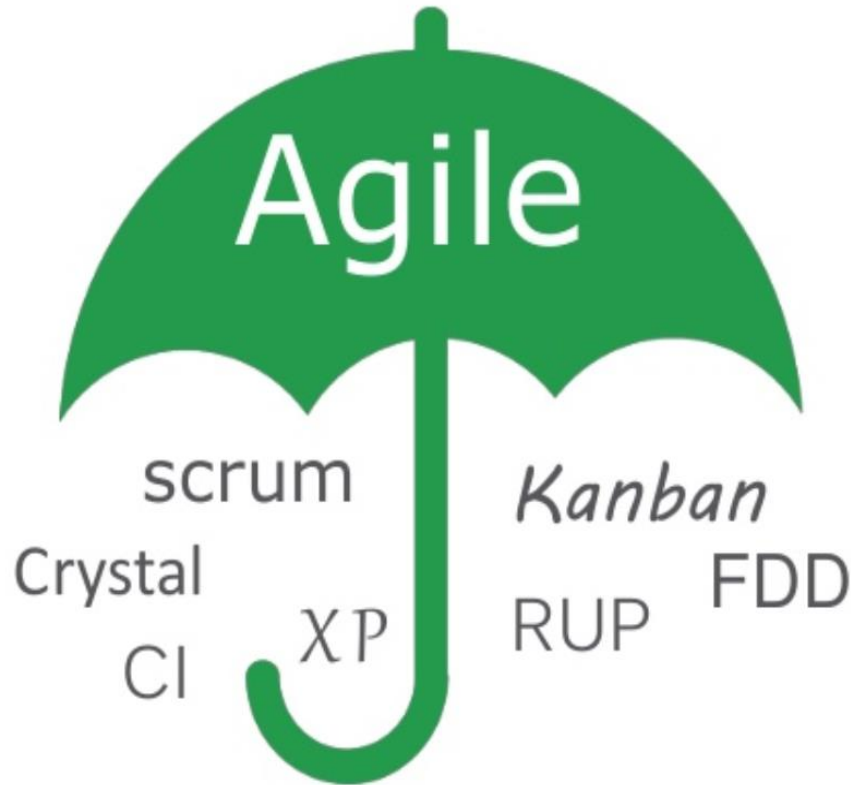
Agile Manifesto

Customer Collaboration
Over
Contract Negotiation

Agile Manifesto

Respond to Change
Over
Following a Plan

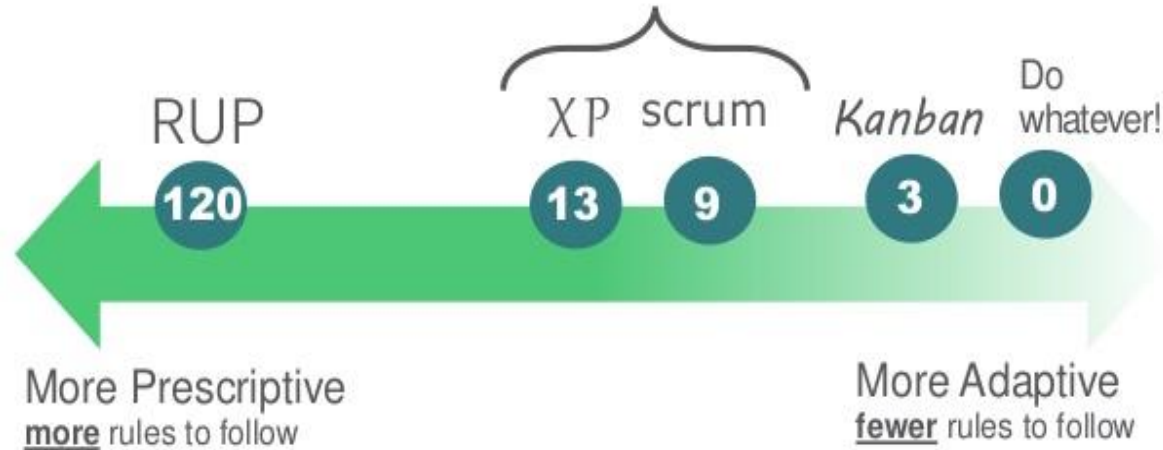
Agile Methodologies



Agile Methodologies

Prescriptive vs. Adaptive

The Sweet Spot



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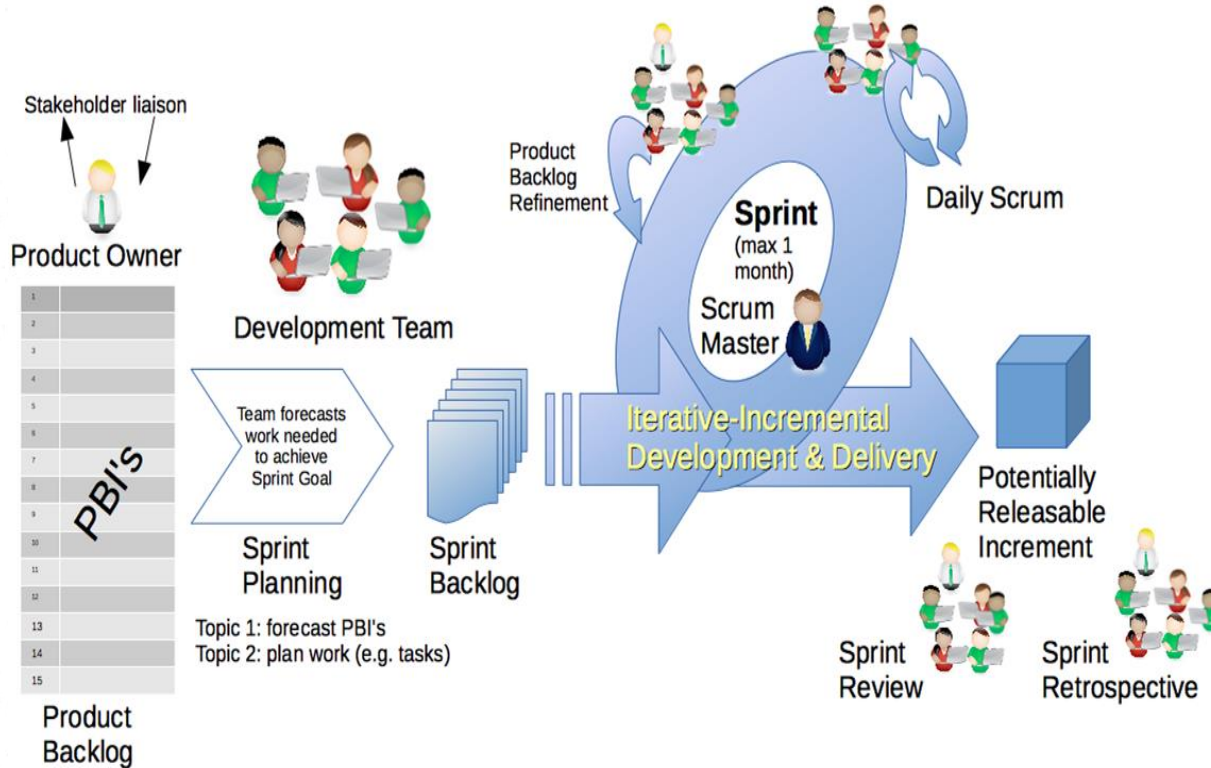
SCRUM Master

Servant Leader
Resolves blockers
Guardian of the Project!

SCRUM Master



SCRUM Framework



Writing User-stories/Epics

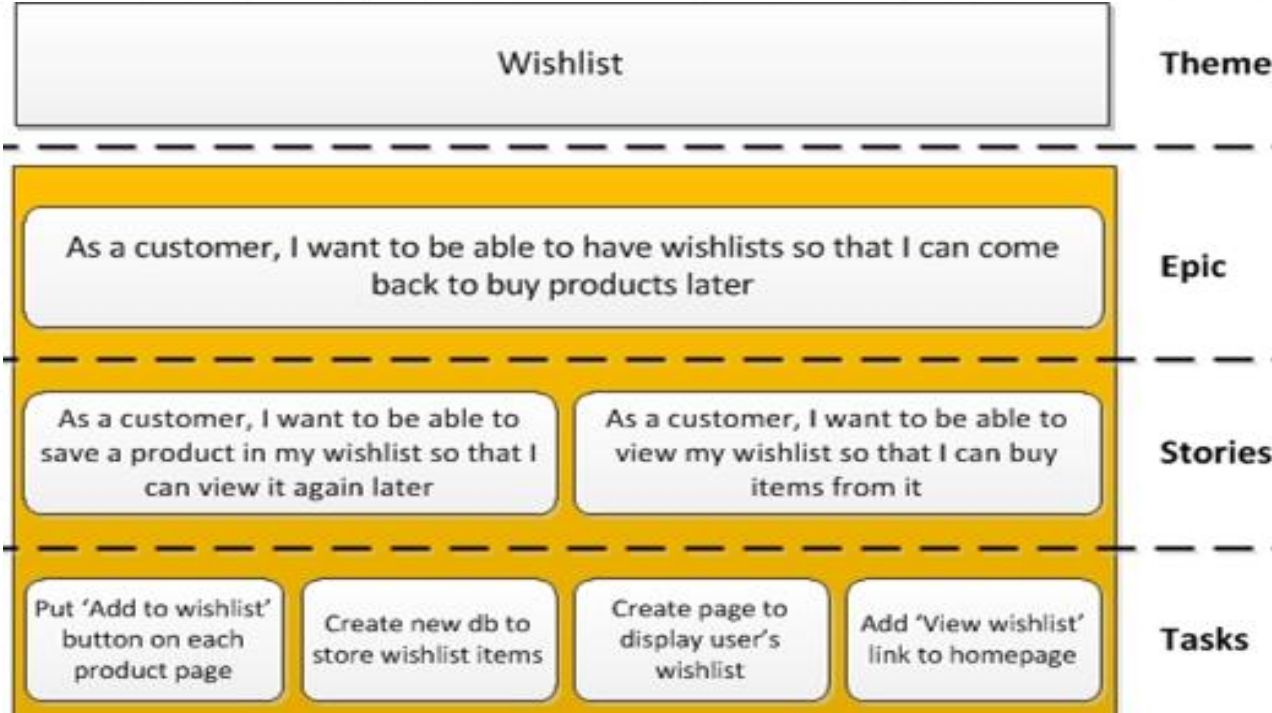


As <persona> ,

I want <what?>

so that <why?>.

Writing User-stories/Epics



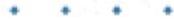
Backlog Grooming

- Removing user stories that no longer appear relevant
- Creating new user stories in response to newly discovered needs
- Re-assessing the relative priority of stories
- Assigning estimates to stories which have yet to receive one
- Correcting estimates in light of newly discovered information
- Splitting user stories which are high priority but too coarse grained to fit in an upcoming iteration

Backlog Grooming

How its Done!
(Video)

Scrum Estimation



Planning Poker



Agile Planning Poker Cards

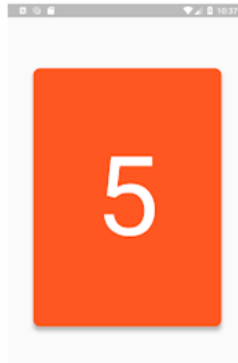
Sam Carton Tools

★★★★★ 17

3+

This app is compatible with some of your devices.

Installed



<https://play.google.com/store/apps/details?id=com.samcarton.planningpoker&hl=en>



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Planning Poker - Steps

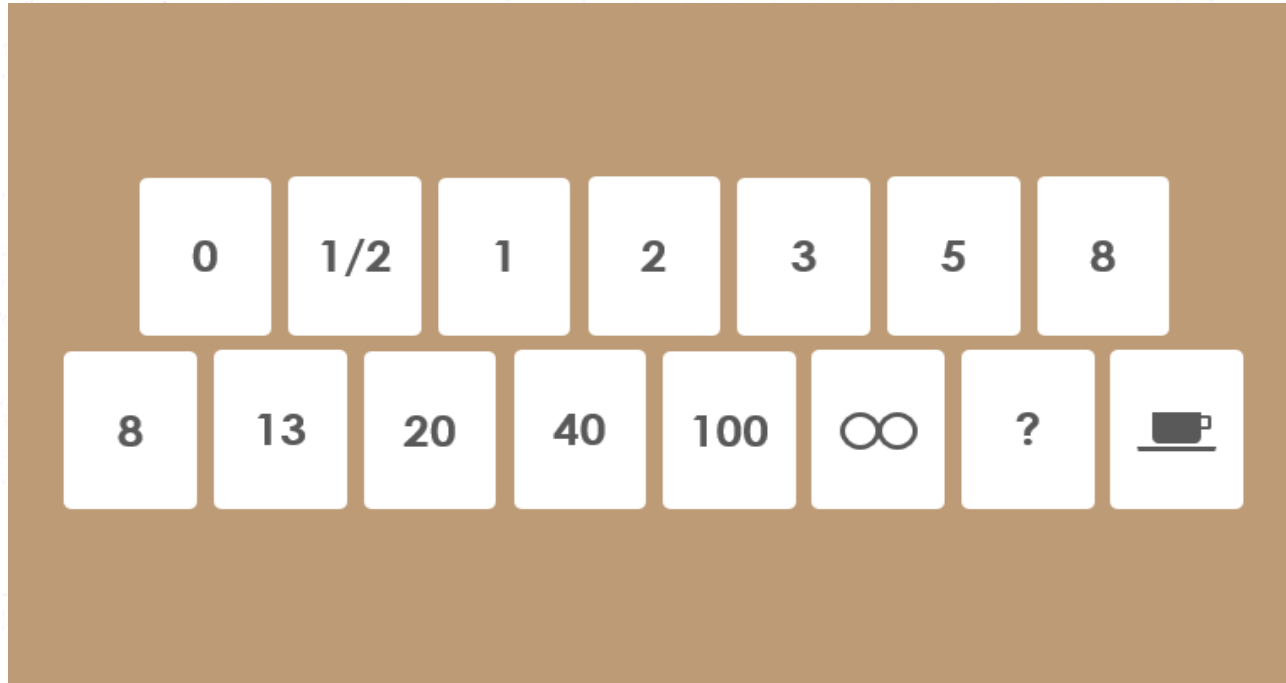


1. The entire scrum team gathers with planning poker cards or a planning poker mobile application.
2. Scrum master takes each item in the product backlog and discusses it with the team.
3. Each team member individually selects a value from the cards or from the mobile application. It is not revealed to anyone yet.
4. When the scrum master gives the signal, everyone reveals the estimated values.

Planning Poker - Steps

5. After looking at all the values, the members with the largest values and the smallest values explain the reasons behind their estimations. Everyone listens to the explanations and if needed others can add their comments as well.
6. Next, they go for another round in the same manner.
7. This is performed until the estimates converge to a closer number or majority of the team members votes for the same value.
8. The next backlog item is selected, and the same process is performed.
9. This is done until the team feels that they have estimated sufficient items to fill in for a new sprint.

Planning Poker



Planning Poker

How its Done! (Video)

Planning poker cards typically feature values from the Fibonacci sequence (1, 2, 3, 5, 8, 13, etc.) or T-shirt sizes (XS, S, M, L, XL) to estimate the relative effort or complexity of tasks in Agile project planning. These values help team members provide estimates during planning sessions.

Daily Scrum

What did you do yesterday?
What are you going to do today?
What are your Impediments?

Daily Scrum

How its Done!
(Video)

Thank You



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Project Selection



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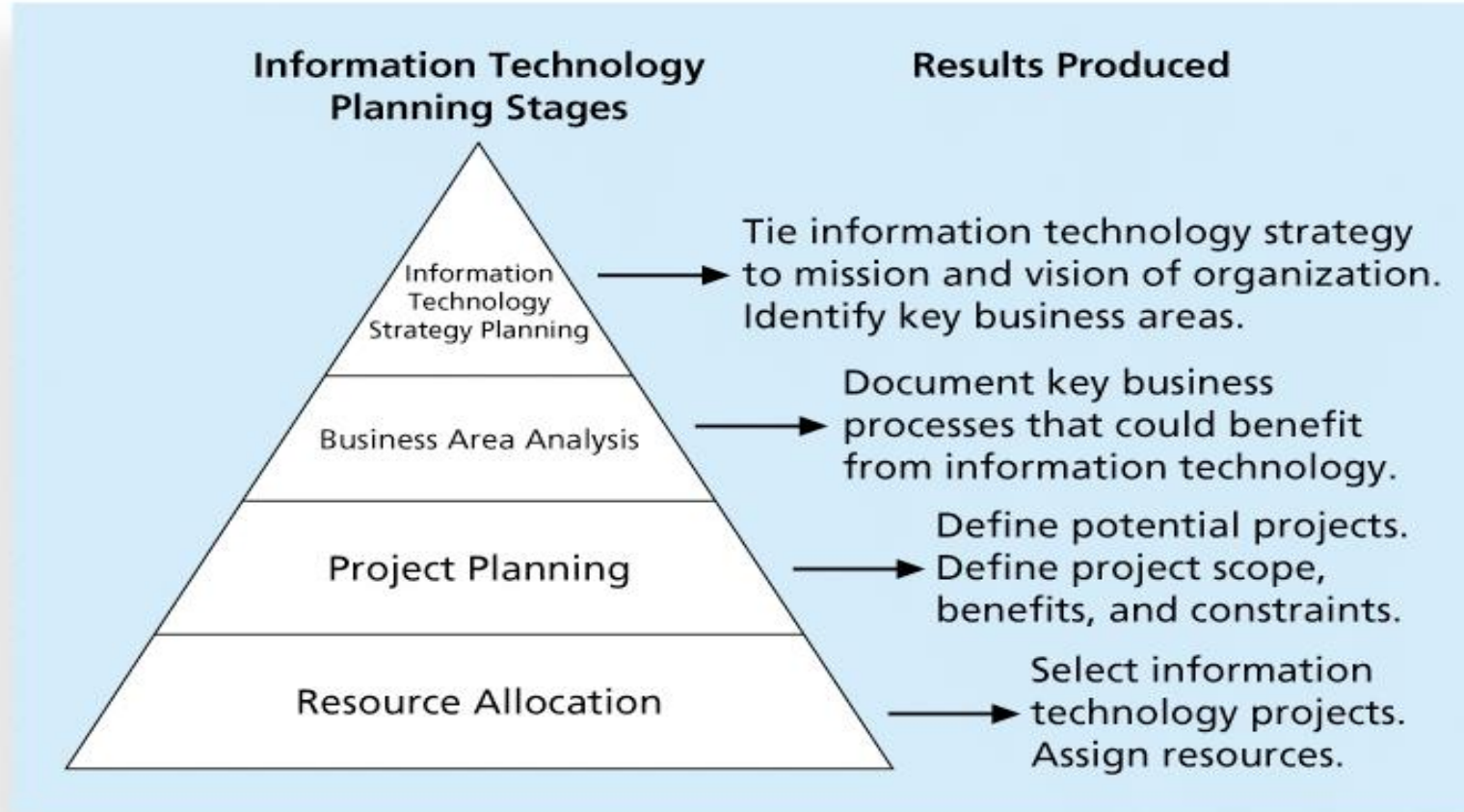


Strategic Planning & Project Selection



- Strategic planning involves determining long-term objectives, predicting future trends, and projecting the need for new products and services.
- Organizations often perform a SWOT analysis
 - Analyzing Strengths, Weaknesses, Opportunities, and Threats
- As part of strategic planning, organizations
 - Identify potential projects
 - Use realistic methods to select which projects to work on
 - Formalize project initiation by issuing a project charter

Information Technology Planning Process





Methods for Selecting Projects

- There are usually more projects than available time and resources to implement them.
- Methods for selecting projects include:
 - focusing on broad organizational needs
 - categorizing information technology projects
 - performing net present value or other financial analyses
 - using a weighted scoring model
 - implementing a balanced scorecard



Focusing on Broad Organizational Needs



- It is often difficult to provide strong justification for many IT projects, but everyone agrees they have a high value.
- “It is better to measure gold roughly than to count pennies precisely”
- Three important criteria for projects:
 - There is a need for the project
 - There are funds available
 - There’s a strong will to make the project succeed



Categorizing IT Projects



- One categorization is whether the project addresses
 - A problem
 - An opportunity, or
 - A directive
- Another categorization is how long it will take to do and when it is needed.
- Another is the overall priority of the project.



Financial Analysis of Projects

- Financial considerations are often an important consideration in selecting projects.
- Three primary methods for determining the projected financial value of projects:
 - Net present value (NPV) analysis
 - Return on investment (ROI)
 - Payback analysis



Net Present Value Analysis

- Net present value (NPV) analysis is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time.
- Projects with a positive NPV should be considered if financial value is a key criterion.
- The higher the NPV, the better

Net Present Value Example

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV →	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV →	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money



NPV Calculations

- Determine estimated costs and benefits for the life of the project and the products it produces.
- Determine the discount rate (check with your organization on what to use).
- Calculate the NPV
- Notes: Some organizations consider the investment year as year 0, while others start in year 1. Some people entered costs as negative numbers, while others do not. Check with your organization for their preferences



Return on Investment



- Return on investment (ROI) is calculated by subtracting the project costs from the benefits and then dividing by the costs

- $$\text{ROI} = (\text{total discounted benefits} - \text{total discounted costs}) / \text{discounted costs}$$

This tells you how much money you will get back compared to how much you spend on the project.

- The higher the ROI, the better.
- Many organizations have a required rate of return or minimum acceptable rate of return on investment for projects.
- Internal rate of return (IRR) can be calculated by finding the discount rate that makes the NPV equal to zero.



Payback Analysis



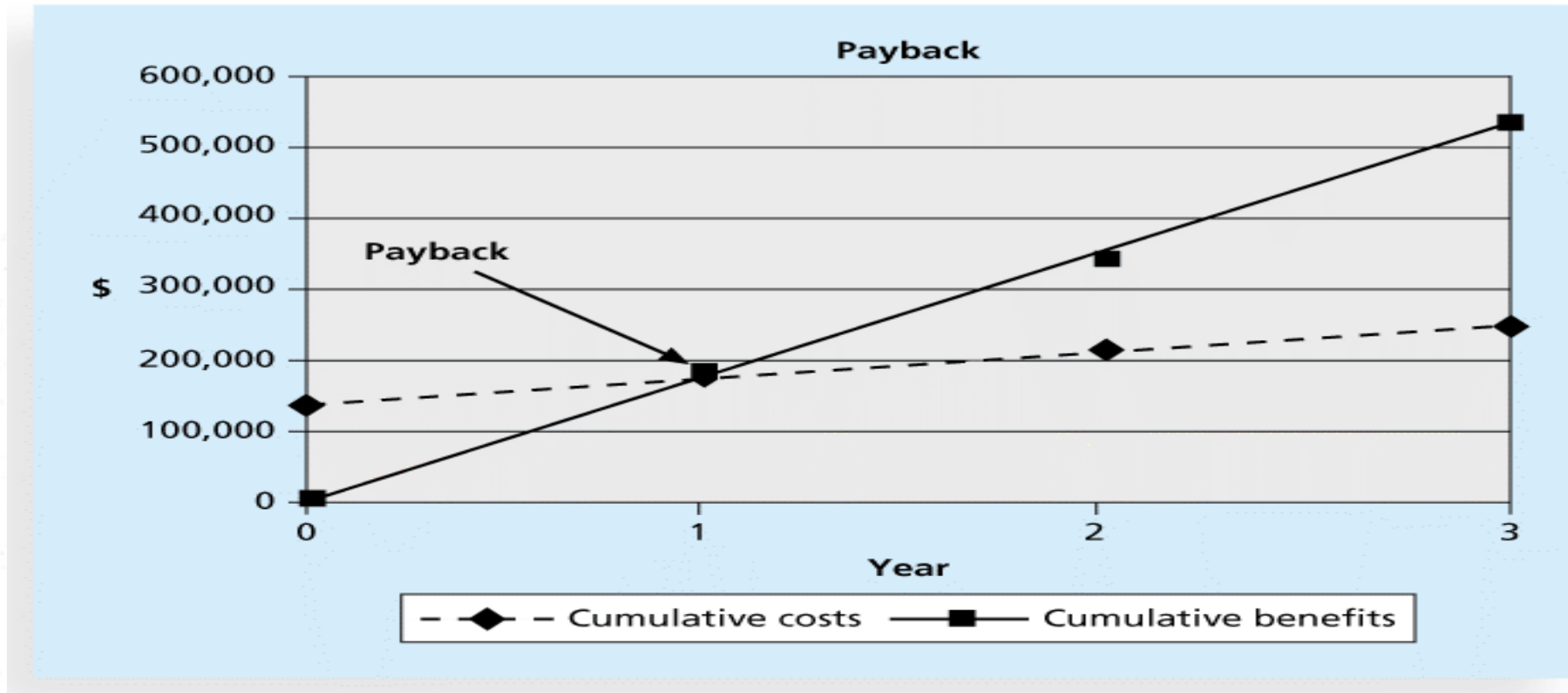
- Another important financial consideration is payback analysis.
- The payback period is the amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project.
- Payback occurs when the net cumulative discounted benefits equals the costs.
- Many organizations want IT projects to have a fairly short payback period.

This shows how long it will take to get back the money you spent on the project.

NPV Example

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	0	186,000	172,000	158,000	516,000	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	112%					
	Payback In Year 1					

Charting the Payback Period





Weighted Scoring Model

- A weighted scoring model is a tool that provides a systematic process for selecting projects based on many criteria
 - Identify criteria important to the project selection process
 - Assign weights (percentages) to each criterion so they add up to 100%
 - Assign scores to each criterion for each project
 - Multiply the scores by the weights and get the total weighted scores
- The higher the weighted score, the better.

Sample Weighted Scoring Model for Project Selection

	A	B	C	D	E	F
1	Criteria	Weight	Project 1	Project 2	Project 3	Project 4
2	Supports key business objectives	25%	90	90	50	20
3	Has strong internal sponsor	15%	70	90	50	20
4	Has strong customer support	15%	50	90	50	20
5	Uses realistic level of technology	10%	25	90	50	70
6	Can be implemented in one year or less	5%	20	20	50	90
7	Provides positive NPV	20%	50	70	50	50
8	Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
9	Weighted Project Scores	100%	56	78.5	50	41.5
10						
11						
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21						
22						
23						
24						
25						
26						

Weighted Score by Project

Project	Weighted Score
Project 4	41.5
Project 3	50
Project 2	78.5
Project 1	56



Implementing a Balanced Scorecard



- Drs. Robert Kaplan and David Norton developed this approach to help select and manage projects that align with business strategy.
 - A balanced scorecard
 - Is a methodology that converts an organization's value drivers, such as customer service, innovation, operational efficiency, and financial performance, to a series of defined metrics.
- 

Balanced Scorecard Example

Mission: Provide responsive, professional finance and accounting services for the people who defend America

Vision:

Best Value to our customers

- World-class provider of finance and accounting services
- Trusted, innovative financial partner
- One Organization, One Identity
- Employer of choice, providing a progressive and professional work environment

Goals

- Fully satisfy customer requirements and aggressively resolve problems to deliver best value services
- Use performance metrics to drive best business practices and achieve high quality results
- Optimize the mix of our military, civilian, and contractor workforce
- Establish consultative relationships with leaders
- Deliver business intelligence to enable better decisions
- Ensure everyone is working towards the same vision and can connect what they're doing to make that vision a reality
- Embrace continuous learning for our workforce to ensure critical, high quality skill sets
- Develop the next generation of DFAS leadership

CUSTOMER PERSPECTIVE

- Improve client/customer satisfaction

FINANCIAL PERSPECTIVE

- Reduce cost to the client/customer
- Expand the use of competitive sourcing

INTERNAL PERSPECTIVE

- Improve and leverage quality
- Encourage innovation
- Deliver system solutions

GROWTH & LEARNING PERSPECTIVE

- Enhance employee competence
- Increase employee satisfaction
- Enhance ability to recruit and retain DFAS talent
- Develop climate for action



The Key to Overall Project Success: Good Project Integration Management



- Project managers must coordinate all of the other knowledge areas throughout a project's life cycle.
- Many new project managers have trouble looking at the “big picture” and want to focus on too many details.
- Project integration management is not the same thing as software integration



Project Integration Management Processes

1. Developing the project charter involves working with stakeholders to create the document that formally authorizes a project—the charter.
2. Developing the project management plan involves coordinating all planning efforts to create a consistent, coherent document—the project management plan.
3. Directing and managing project work involves carrying out the project management plan by performing the activities included in it.



Project Integration Management Processes (cont'd)



4. Monitoring and controlling project work involves overseeing activities to meet the performance objectives of the project
5. Performing integrated change control involves identifying, evaluating, and managing changes throughout the project life cycle.
6. Closing the project or phase involves finalizing all activities to formally close the project or phase.

Project Integration Management Summary

Influencing the Factors that Create Changes:

What it means: Try to make sure that any changes to the project are good and helpful.
How it's done: By planning carefully and communicating well, you can guide changes in a positive direction.
Determining that a Change Has Occurred:

What it means: Notice when something in the project has changed.
How it's done: By keeping a close watch on the project activities and comparing them to the plan, you can spot any changes that happen.
Managing Actual Changes as They Occur:

What it means: Handle the changes when they happen to make sure they don't mess up the project.
How it's done: By following a process to evaluate and approve changes, you can manage them effectively and keep the project on track.

Initiating

Process: **Develop project charter**

Output: Project charter

Planning

Process: **Develop project management plan**

Output: Project management plan

Executing

Process: **Direct and manage project work**

Outputs: Deliverables, work performance data, change requests, project management plan updates, project documents updates

Monitoring and Controlling

Process: **Monitor and control project work**

Outputs: Change requests, project management plan updates, project documents updates

Process: **Perform integrated change control**

Outputs: Approved change requests, change log, project management plan updates, project documents updates

Closing

Process: **Close project or phase**

Outputs: Final product, service, or result transition; organizational process assets updates

Project Start

Project Finish



Performing Integrated Change Control



- Three main objectives are:
 - Influencing the factors that create changes to ensure that changes are beneficial
 - Determining that a change has occurred
 - Managing actual changes as they occur
- 



Change Control on Information Technology Projects



- Former view: The project team should strive to do exactly what was planned on time and within budget.
- Problem: Stakeholders rarely agreed up-front on the project scope, and time and cost estimates were inaccurate.
- Modern view: Project management is a process of constant communication and negotiation.
- Solution: Changes are often beneficial, and the project team should plan for them.



Change Control System

- A change control system is a formal, documented process that describes when and how official project documents and work may be changed.
- Describes who is authorized to make changes and how to make them



Change Control Board (CCB)

- A change control board is a formal group of people responsible for approving or rejecting changes on a project.
- CCBs provide guidelines for preparing change requests, evaluate change requests, and manage the implementation of approved changes.
- Includes stakeholders from the entire organization.



Making Timely Changes



- Some CCBs only meet occasionally, so it may take too long for changes to occur.
- Some organizations have policies in place for time-sensitive changes:
 - “48-hour policy” allows project team members to make decisions, then they have 48 hours to reverse the decision pending senior management approval
 - Delegate changes to the lowest level possible, but keep everyone informed of changes



Configuration Management

- Configuration management ensures that the descriptions of the project's products are correct and complete.
- Involves identifying and controlling the functional and physical design characteristics of products and their support documentation.
- Configuration management specialists identify and document configuration requirements, control changes, record and report changes, and audit the products to verify conformance to requirements.



Suggestions for Performing Integrated Change Control



View project management as a process of constant communication and negotiation.

Plan for change.

- Establish a formal change control system, including a change control board (CCB).
 - • Use effective configuration management.
 - • Define procedures for making timely decisions on smaller changes.
 - • Use written and oral performance reports to help identify and manage change.
 - • Use project management and other software to help manage and communicate changes.
-

Focus on leading the project team and meeting overall project goals and expectations.

Project Cost Management

Learning Objectives

- Understand the importance of project cost management.
- Explain basic project cost management principles, concepts, and terms.
- Describe the process of planning cost management.
- Discuss different types of cost estimates and methods for preparing them.

Learning Objectives

- Understand the processes of determining a budget and preparing a cost estimate for an information technology (IT) project.
- Understand the benefits of earned value management.

The Importance of Project Cost Management

- IT projects have a poor track record for meeting budget goals.
- The CHAOS studies found the average cost overrun (the additional percentage or dollar amount by which actual costs exceed estimates) ranged from 180 percent in 1994 to 45 percent in 2001.
- A 2011 Harvard Business Review study reported an average cost overrun of 27 percent. The most important finding was the discovery of a large number of gigantic overages or “black swans”.

What is Cost and Project Cost Management?



- Cost is a resource sacrificed or foregone to achieve a specific objective or something given up in exchange.
- Costs are usually measured in monetary units like dollars.
- Project cost management includes the processes required to ensure that the project is completed within an approved budget.

Project Cost Management Processes

- **Planning cost management:** Determining the policies, procedures, and documentation that will be used for planning, executing, and controlling project cost.
- **Estimating costs:** Developing an approximation or estimate of the costs of the resources needed to complete a project.
- **Determining the budget:** Allocating the overall cost estimate to individual work items to establish a baseline for measuring performance.
- **Controlling costs:** Controlling changes to the project budget.

Project Cost Management Summary

Planning

Process: **Plan cost management**

Outputs: Cost management plan

Process: **Estimate costs**

Outputs: Activity cost estimates, basis of estimates, project documents updates

Process: **Determine budget**

Outputs: Cost baseline, project funding requirements, project documents updates

Monitoring and Controlling

Process: **Control costs**

Outputs: Work performance information, cost forecasts, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish

Basic Principles of Cost Management

- Most members of an executive board better understand and are more interested in financial terms than IT terms, so IT project managers must speak their language:
 - Profits are revenues minus expenditures
 - Profit margin is the ratio of revenues to profits
 - Life cycle costing considers the total cost of ownership, or development plus support costs, for a project
 - Cash flow analysis determines the estimated annual costs and benefits for a project and the resulting annual cash flow

Types of Costs and Benefits

- **Tangible costs** or benefits are those costs or benefits that an organization can easily measure in dollars.
- **Intangible costs** or benefits are costs or benefits that are difficult to measure in monetary terms.
- **Direct costs** are costs that can be directly related to producing the products and services of the project.
- **Indirect costs** are costs that are not directly related to the products or services of the project but are indirectly related to performing the project.
- **Sunk cost** is money that has been spent in the past; when deciding what projects to invest in or continue, you should not include sunk costs.

More Basic Principles of Cost Management

- **Learning curve theory** states that when many items are produced repetitively, the unit cost of those items decreases in a regular pattern as more units are produced.
- Reserves are dollars included in a cost estimate to mitigate cost risk by allowing for future situations that are difficult to predict:
 - **Contingency reserves** allow for future situations that may be partially planned for (sometimes called known unknowns) and are included in the project cost baseline.
 - **Management reserves** allow for future situations that are unpredictable (sometimes called unknown unknowns).

Planning Cost Management

- The project team uses expert judgment, analytical techniques, and meetings to develop the cost management plan.
- A cost management plan includes:
 - Level of accuracy and units of measure
 - Organizational procedure links
 - Control thresholds
 - Rules of performance measurement
 - Reporting formats
 - Process descriptions

Estimating Costs

- Project managers must take cost estimates seriously if they want to complete projects within budget constraints.
- It's important to know the types of cost estimates, how to prepare cost estimates, and typical problems associated with IT cost estimates.

Types of Cost Estimates

TYPE OF ESTIMATE	WHEN DONE	WHY DONE	HOW ACCURATE
Rough Order of Magnitude (ROM)	Very early in the project life cycle, often 3–5 years before project completion	Provides estimate of cost for selection decisions	–50% to +100%
Budgetary	Early, 1–2 years out	Puts dollars in the budget plans	–10% to +25%
Definitive	Later in the project, less than 1 year out	Provides details for purchases, estimates actual costs	–5% to +10%

Cost Estimation Tools and Techniques

- Basic tools and techniques for cost estimates:
 - **Analogous or top-down estimates:** use the actual cost of a previous, similar project as the basis for estimating the cost of the current project.
 - **Bottom-up estimates:** involve estimating individual work items or activities and summing them to get a project total.
 - Parametric modeling uses project characteristics (parameters) in a mathematical model to estimate project costs.

Typical Problems with IT Cost Estimates

- Estimates are done too quickly.
- People lack estimating experience.
- Human beings are biased toward underestimation.
- Management desires accuracy.

Determining the Budget

- Cost budgeting involves allocating the project cost estimate to individual work items over time.
- The WBS is a required input to the cost budgeting process since it defines the work items.
- Important goal is to produce a cost baseline:
 - A time-phased budget that project managers use to measure and monitor cost performance.

Controlling Costs

- Project cost control includes:
 - Monitoring cost performance.
 - Ensuring that only appropriate project changes are included in a revised cost baseline.
 - Informing project stakeholders of authorized changes to the project that will affect costs.
- Many organizations around the globe have problems with cost control.

Earned Value Management (EVM)

- **Earned value management (EVM)** is a methodology that combines scope, schedule, and resource measurements to assess project performance and progress.
- **Planned value (PV)** is the authorized budget assigned to scheduled work.
- **Earned value (EV)** is a measure of work performed expressed in terms of the budget authorized for that work.
- **Actual cost (AC)** is the realized cost incurred for the work performed on an activity during a specific time period.

Variance Calculations

- **Schedule Variance (SV)** is a measure of schedule performance expressed as the difference between the earned value and the planned value.

$$SV = EV - PV$$

- It is the amount by which the project is ahead or behind the planned delivery date, at a given point in time. It is a measure of schedule performance on a project.

Variance Calculations cont..

- **Cost variance (CV)** is the amount of budget shortage or excess at a given point in time, expressed as the **difference between earned value and the actual cost**.

$$CV = EV - AC$$

- It is a measure of cost performance on a project.
- **Budget at Completion (BAC)** is the total approved budget for the project.
- Therefore, **Variance at Completion (VAC)** is:
VAC = BAC – AC, where AC is the total amount spent for the project

Performance Calculations

- **Schedule Performance Index (SPI)** is a measure of the schedule efficiency expressed as the ratio of earned value to planned value. It measures how efficiently the project team is using its time.

$$SPI = EV / PV$$

Performance Calculations cont..

- **Cost Performance Index (CPI)** is a measure of the cost efficiency of budgeted resources, expressed as a ratio of earned value to actual cost.

$$\text{CPI} = \text{EV} / \text{AC}$$

- It is considered the most critical EVM metric and measures the cost efficiency for the work completed.

Forecasting

- As the project progresses, the project team may develop a forecast for the **Estimate at Completion (EAC)** that may differ from the **Budget at Completion (BAC)** based on the project performance.

$$\text{EAC} = \text{AC} + \text{ETC}$$

Where ETC is estimate to complete

Forecasting cont..

- EAC forecast for ETC work performed at the budgeted rate:

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$$

- EAC forecast for ETC work performed at the present CPI:

$$\text{EAC} = \text{BAC} / \text{CPI}$$

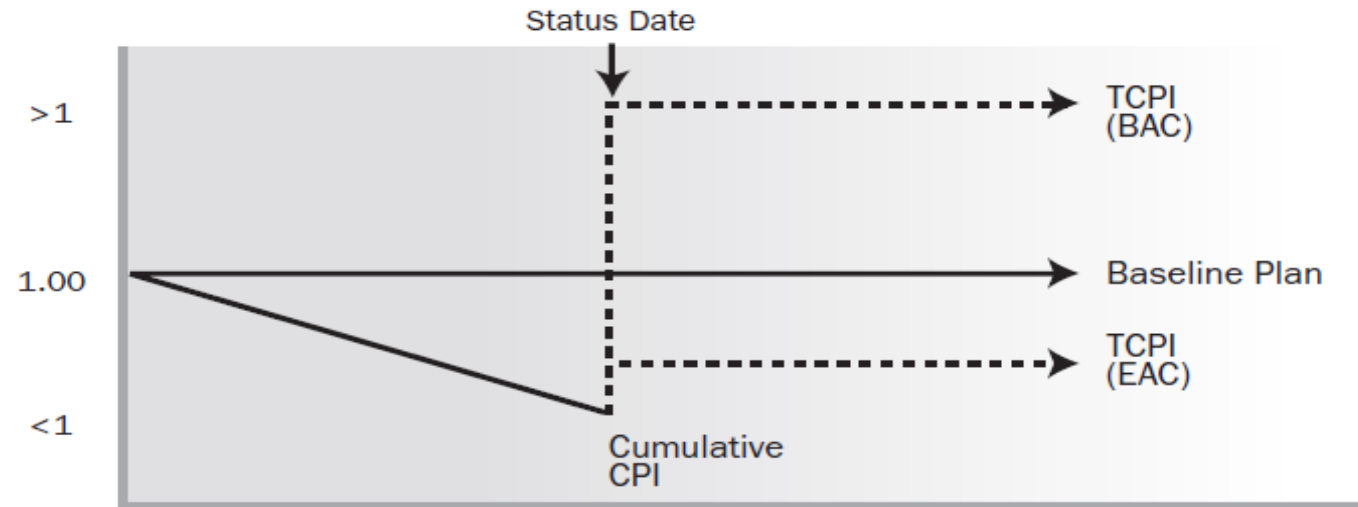
- EAC forecast for ETC work considering both SPI and CPI factors:

$$\text{EAC} = \text{AC} + [(\text{BAC} - \text{EV}) / (\text{CPI} \times \text{SPI})]$$

Forecasting cont..

- **To-Complete Performance Index (TCPI)** is a measure of the cost performance that is required to be achieved with the remaining resources in order to meet a specified management goal, expressed as the ratio of the cost to finish the outstanding work to the remaining budget.

$$\text{TCPI} = \text{Work Remaining} / \text{Funds Remaining}$$



Formula:

$$\frac{\text{Work Remaining (BAC-EV)}}{\text{Funds Remaining (BAC-AC) or (EAC-AC)}} = \text{TCPI}$$

Forecasting cont..

- A TCPI value equal to 1.0 means you are right on budget, and as long as you maintain the present cost performance index (CPI), you should not exceed the project budget.
- A TCPI value greater than 1.0 means you need to take a stricter cost management approach, or you will exceed your budget at completion (BAC).
- A TCPI value less than 1.0 indicates you are within your budget and have done an excellent job of managing project costs.
- **Estimated Duration to Complete is:**

$$\text{Estimated Duration to Complete} = \text{Original Duration} / \text{SPI}$$

Exercise

- An organization has spent a total expenditure to date of 15,000,000 from its total budget of 100,000,000. They have completed 40% of the work till now. The budgeted cost of the work scheduled for the current date is 10,000,000.
- Find the following:
 - Cost Variance (CV)
 - Schedule Variance (SV)
 - Variance at Completion (VAC)
 - Estimate at Completion (EAC)

Exercise - Solution

- According to the given data, the project has an earned value of 40% out of the entire budget at completion(BAC). Thus;

$$EV = 40\% * BAC.$$

$$EV = 0.4 * 100,000,000$$

$$EV = 40,000,000$$

- They have spent 15,000,000 of an Actual Cost (AC) to complete this 40% of work. Thus;

$$CV = EV - AC$$

$$CV = 40,000,00 - 15,000,000$$

$$CV = 25,000,000$$

Exercise – Solution cont..

- The Planned Value is the budgeted scheduled cost for the current date which has been given as 10,000,000. Thus;
 - $SV = EV - PV$
 $SV = 40,000,000 - 10,000,000 = 30,000,000$
 - $VAC = BAC - AC$
 $VAC = 100,000,000 - 15,000,000 = 85,000,000$
 - $EAC = AC + (BAC - EV)$
 $EAC = 15,000,000 + (100,000,000 - 40,000,000) = 75,000,000$

Chapter Summary

- Project cost management is a traditionally weak area of IT projects, and project managers must work to improve their ability to deliver projects within approved budgets.
- Main processes include:
 - Plan cost management
 - Estimate costs
 - Determine the budget
 - Control costs



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Project Quality Management



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Learning Objectives

- Understand the importance of project quality management for information technology (IT) products and services.
- Define project quality management and understand how quality relates to various aspects of IT projects.
- Describe quality management planning and how quality and scope management are related.

Learning Objectives

- Discuss the importance of quality assurance.
- Explain the main outputs of the quality control process.
- Understand the tools and techniques for quality control, such as the Seven Basic Tools of Quality, statistical sampling, Six Sigma, and testing.

Learning Objectives

- Summarize the contributions of noteworthy quality experts to modern quality management.
- Describe how leadership, the cost of quality, organizational influences, expectations, cultural differences, and maturity models relate to improving quality in IT projects.

What Went Wrong?

- In 1986, two hospital patients died after receiving fatal doses of radiation from a Therac 25 machine after a software problem caused the machine to ignore calibration data.
- In one of the biggest software errors in banking history, Chemical Bank mistakenly deducted about \$15 million from more than 100,000 customer accounts

What Went Wrong?

- In August 2008, the Privacy Rights Clearinghouse stated that more than 236 million data records of U.S. residents have been exposed due to security breaches since January 2005.
- In March 2012, Consumer Reports listed several recalls on its Web site in less than 10 days, including LED lights overheating, five different models of cars having problems

What Is Project Quality?

- The International Organization for Standardization (ISO) defines quality as “**the degree to which a set of inherent characteristics fulfils requirements**” (ISO9000:2000).
- Other experts define quality based on:
 - Conformance to requirements: The project’s processes and products meet written specifications
 - Fitness for use: A product can be used as it was intended

What Is Project Quality Management?

- Project quality management ensures that the project will satisfy the needs for which it was undertaken.
- Processes include:
 - Planning quality management: Identifying which quality standards are relevant to the project and how to satisfy them; a metric is a standard of measurement
 - Performing quality assurance: Periodically evaluating overall project performance to ensure the project will satisfy the relevant quality standards
 - Performing quality control: Monitoring specific project results to ensure that they comply with the relevant quality standards

Project Quality Management Summary

Planning

Process: **Plan quality management**

Outputs: Quality management plan, process improvement plan, quality metrics, quality checklists, and project documents updates

Executing

Process: **Perform quality assurance**

Outputs: Change requests, project management plan updates, project documents updates, and organizational process asset updates

Monitoring and Controlling

Process: **Perform quality control**

Outputs: Quality control measurements, validated changes, validated deliverables, work performance information, change requests, project management plan updates, project documents updates, and organizational process asset updates

Project Start

Project Finish

Planning Quality

- Implies the ability to anticipate situations and prepare actions to bring about the desired outcome.
- Important to prevent defects by:
 - Selecting proper materials
 - Training and indoctrinating people in quality
 - Planning a process that ensures the appropriate outcome

Scope Aspects of IT Projects

- Functionality is the degree to which a system performs its intended function.
- Features are the system's special characteristics that appeal to users.
- System outputs are the screens and reports the system generates

Scope Aspects of IT Projects

- Performance addresses how well a product or service performs the customer's intended use.
- Reliability is the ability of a product or service to perform as expected under normal conditions.
- Maintainability addresses the ease of performing maintenance on a product

Controlling Quality

- The main outputs of quality control are:
 - Acceptance decisions
 - Rework
 - Process adjustments
- There are Seven Basic Tools of Quality that help in performing quality control

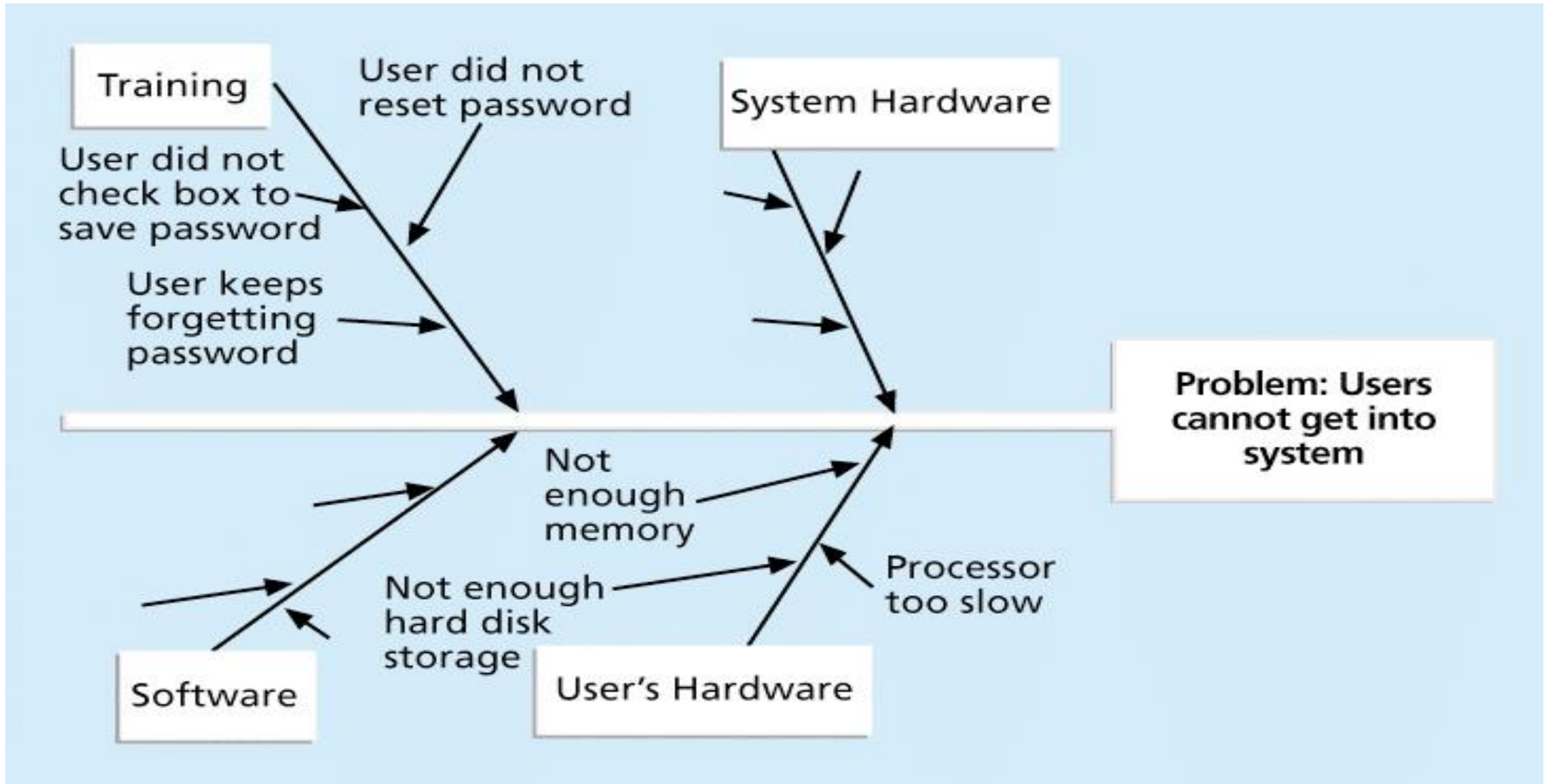
Cause-and-Effect Diagrams

- **Cause-and-effect diagrams** trace complaints about quality problems back to the responsible production operations.

What are causes of issues, how to effect it

- They help you find the root cause of a problem.
- Also known as **fishbone** or **Ishikawa diagrams**.
- Can also use the **5 whys technique** where you repeated ask the question “Why” (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause

Sample Cause-and-Effect Diagram



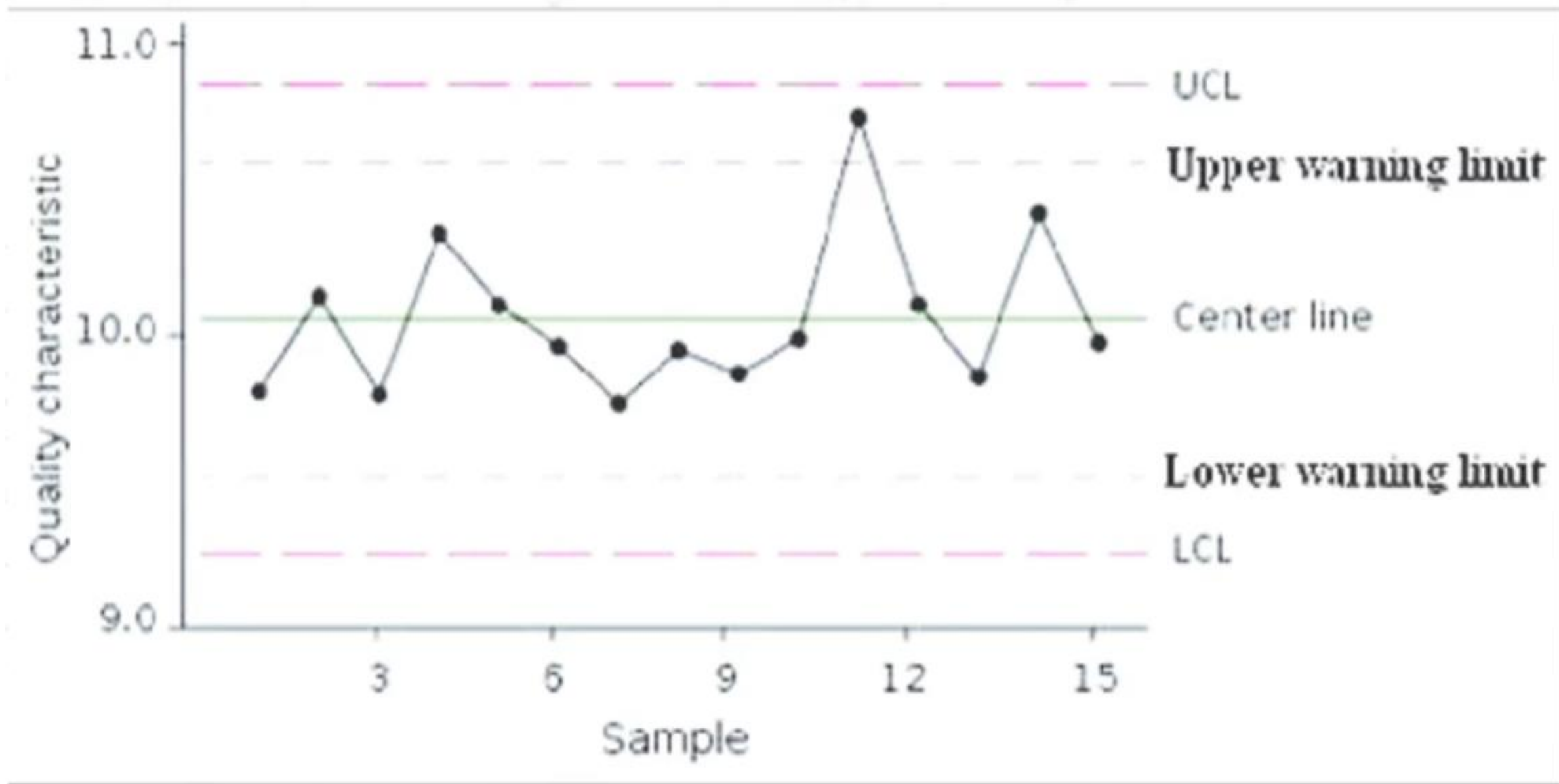
Quality Control Charts

- A control chart is a graphic display of data that illustrates the results of a process over time.
- The main use of control charts is to prevent defects, rather than to detect or reject them.

Quality Control Charts

- Quality control charts allow you to determine whether a process is in control or out of control
 - When a process is in control, any variations in the results of the process are created by random events; processes that are in control do not need to be adjusted
 - When a process is out of control, variations in the results of the process are caused by non-random events; you need to identify the causes of those non-random events and adjust the process to correct or eliminate them

Quality Control Chart



Checksheet

- A checksheet is used to collect and analyze data
- It is sometimes called a tally sheet or checklist, depending on its format
- In the example, most complaints arrive via text message, and there are more complaints on Monday and Tuesday than on other days of the week
- This information might be useful in improving the process for handling complaints

Sample Checksheet

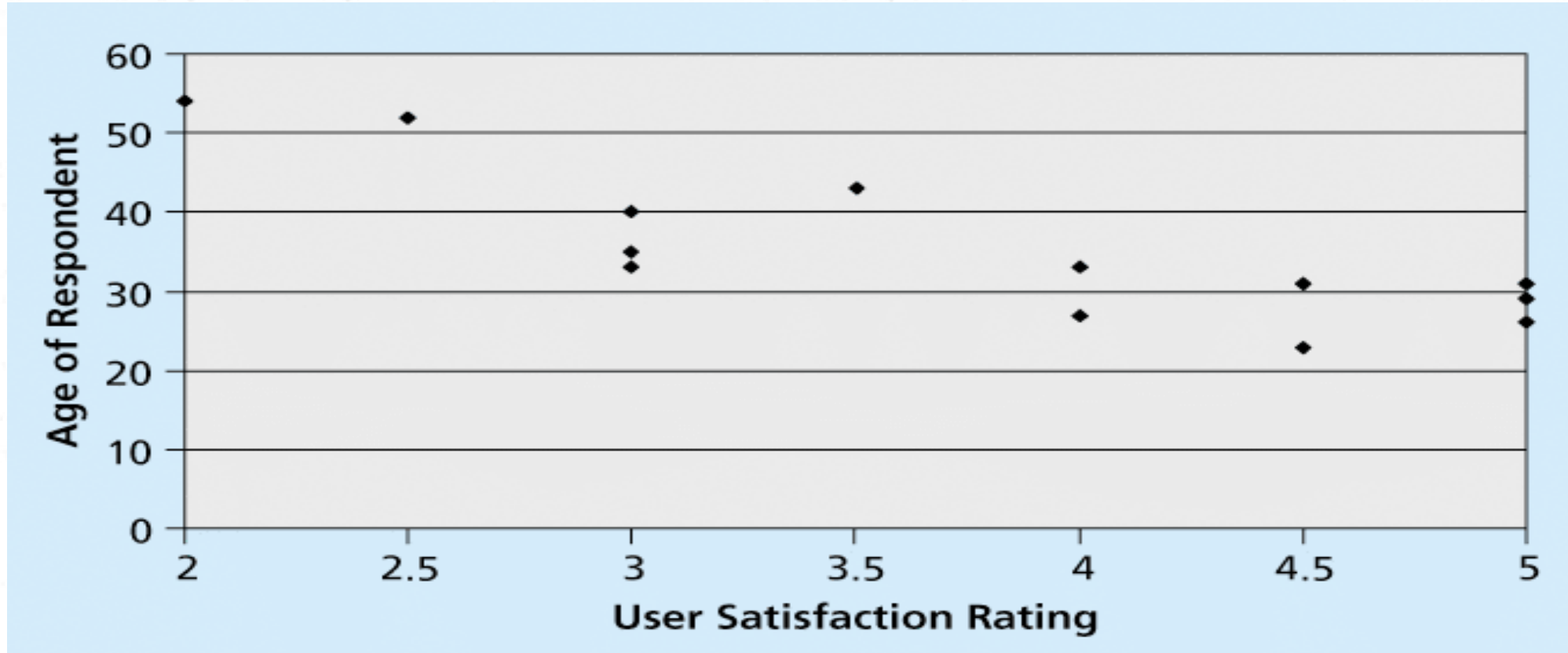
System Complaints

Source	Day							Total
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	
Email								12
Text	 		 					29
Phone call								8
Total	11	10	8	6	7	3	4	49

Scatter Diagram

- A scatter diagram helps to show if there is a relationship between two variables.
- The closer data points are to a diagonal line, the more closely the two variables are related.

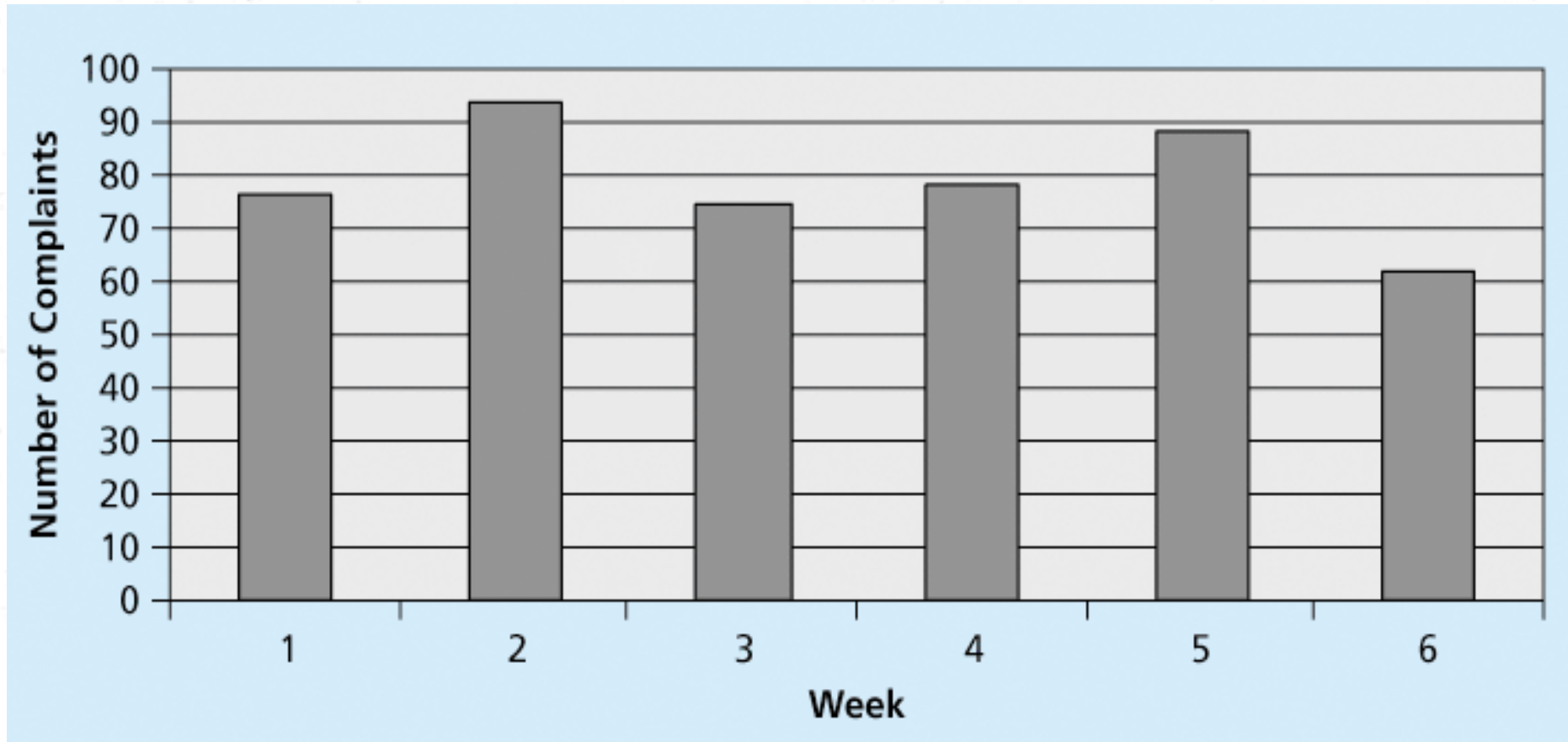
Sample Scatter Diagram



Histograms

- A histogram is a bar graph of a distribution of variables.
- Each bar represents an attribute or characteristic of a problem or situation, and the height of the bar represents its frequency.

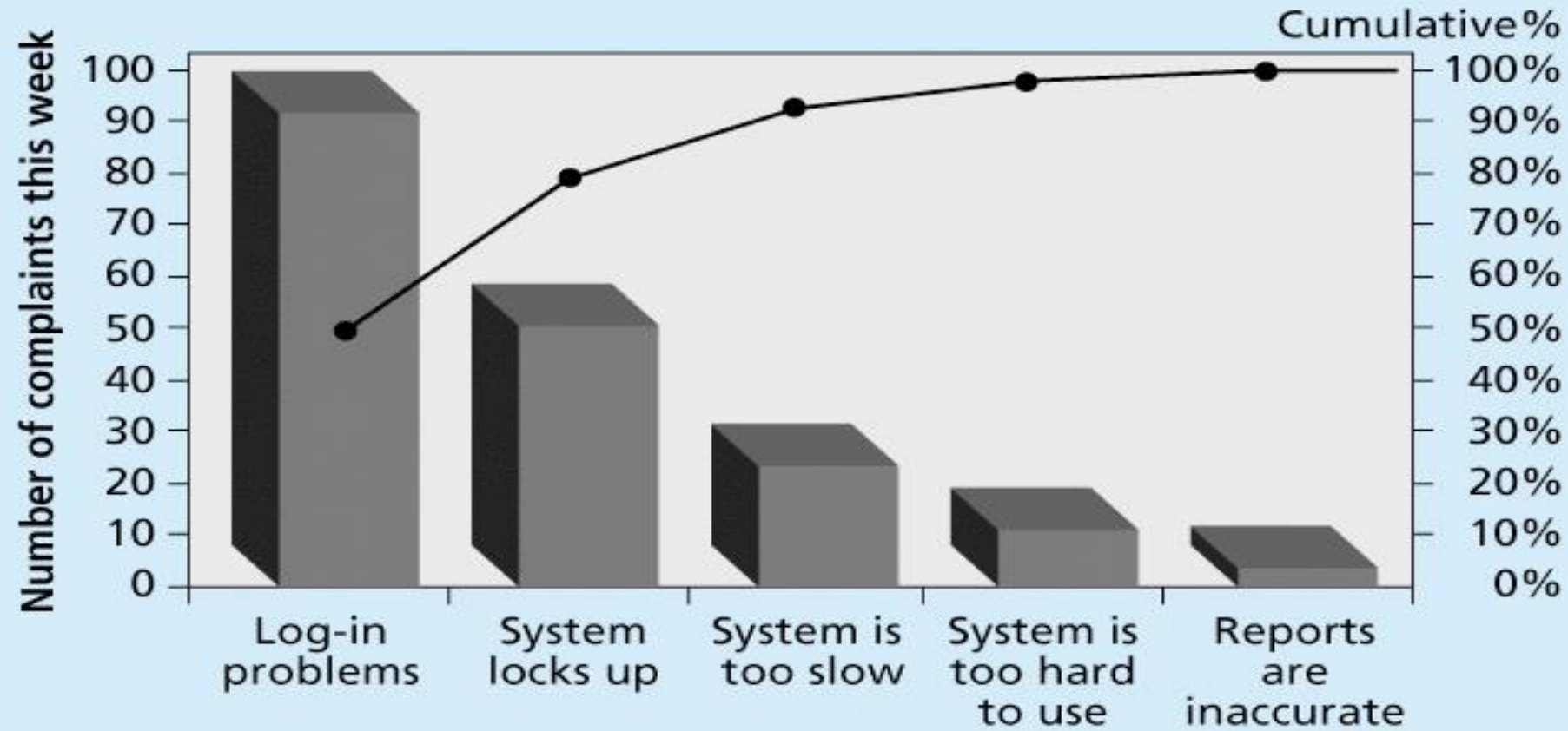
Sample Histogram



Pareto Charts

- A Pareto chart is a histogram that can help you identify and prioritize problem areas.
- Pareto analysis is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes

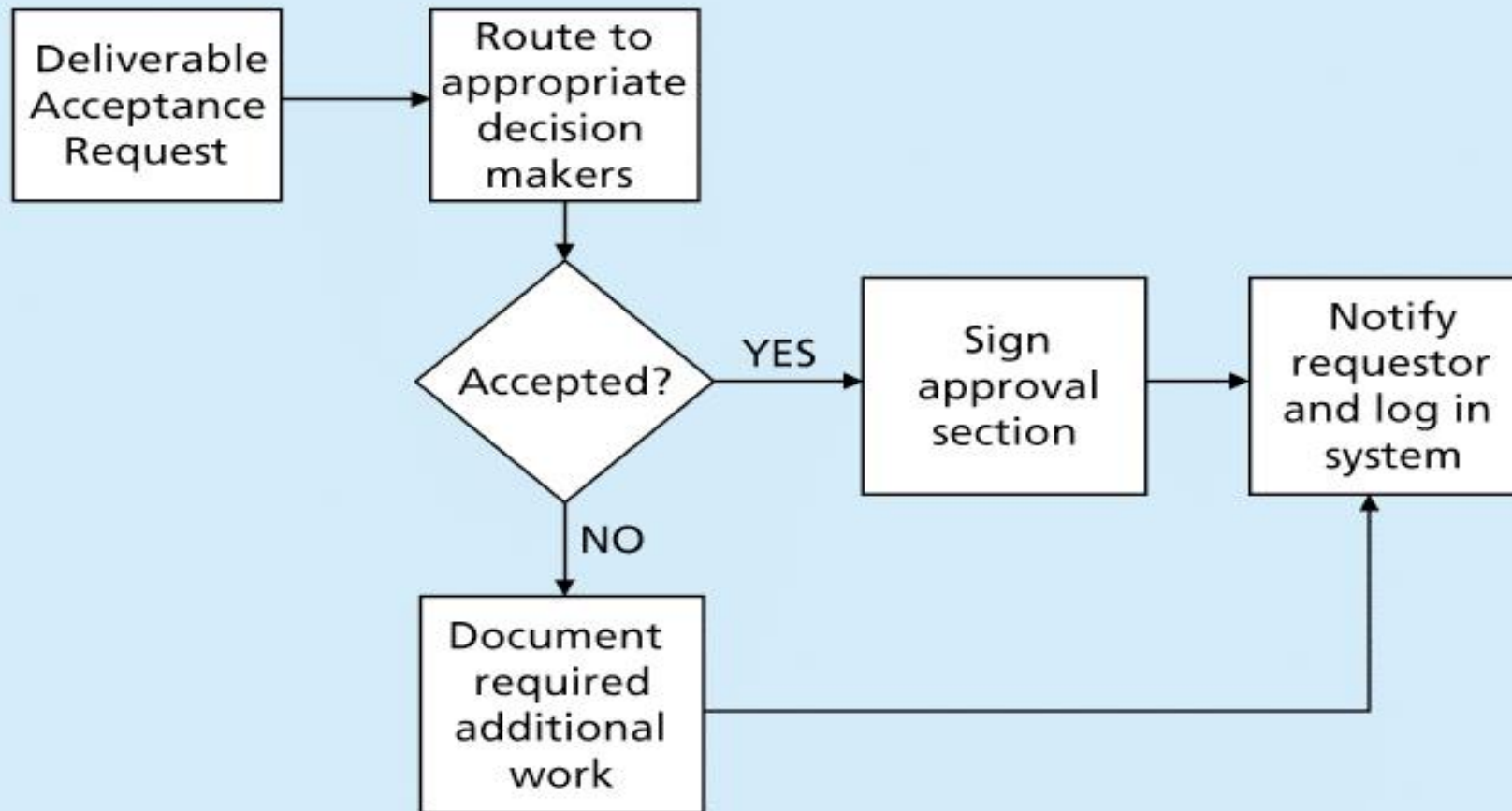
Sample Pareto Chart



Flowcharts

- Flowcharts are graphic displays of the logic and flow of processes that help you analyze how problems occur and how processes can be improved.
- They show activities, decision points, and the order of how information is processed

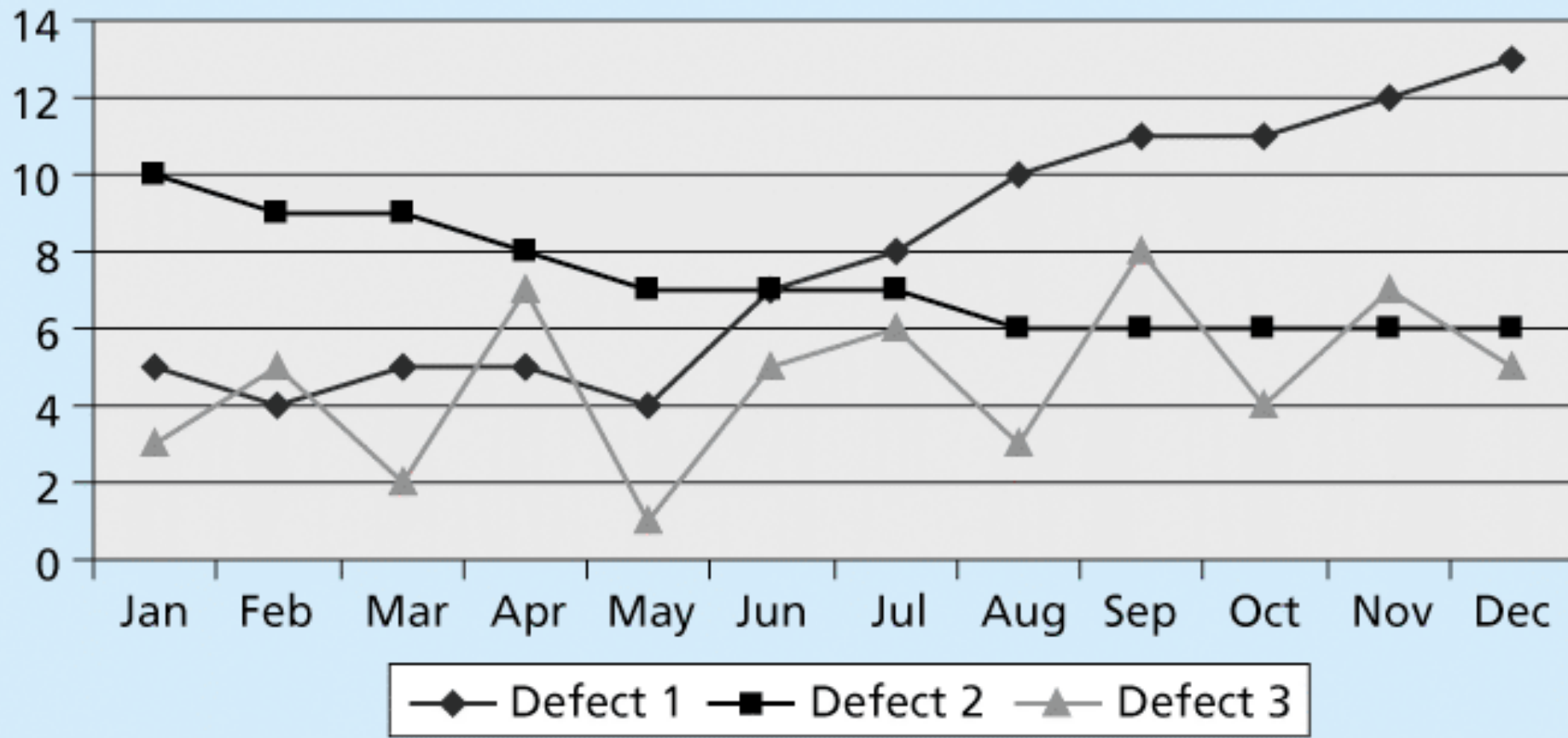
Sample Flowchart



Run Charts

- In addition to flowcharts, run charts are also used for stratification, a technique that shows data from a variety of sources to see if a pattern emerges.
- A run chart displays the history and pattern of variation of a process over time.
- You can use run charts to perform trend analysis and forecast future outcomes based on historical results.

Sample Run Chart



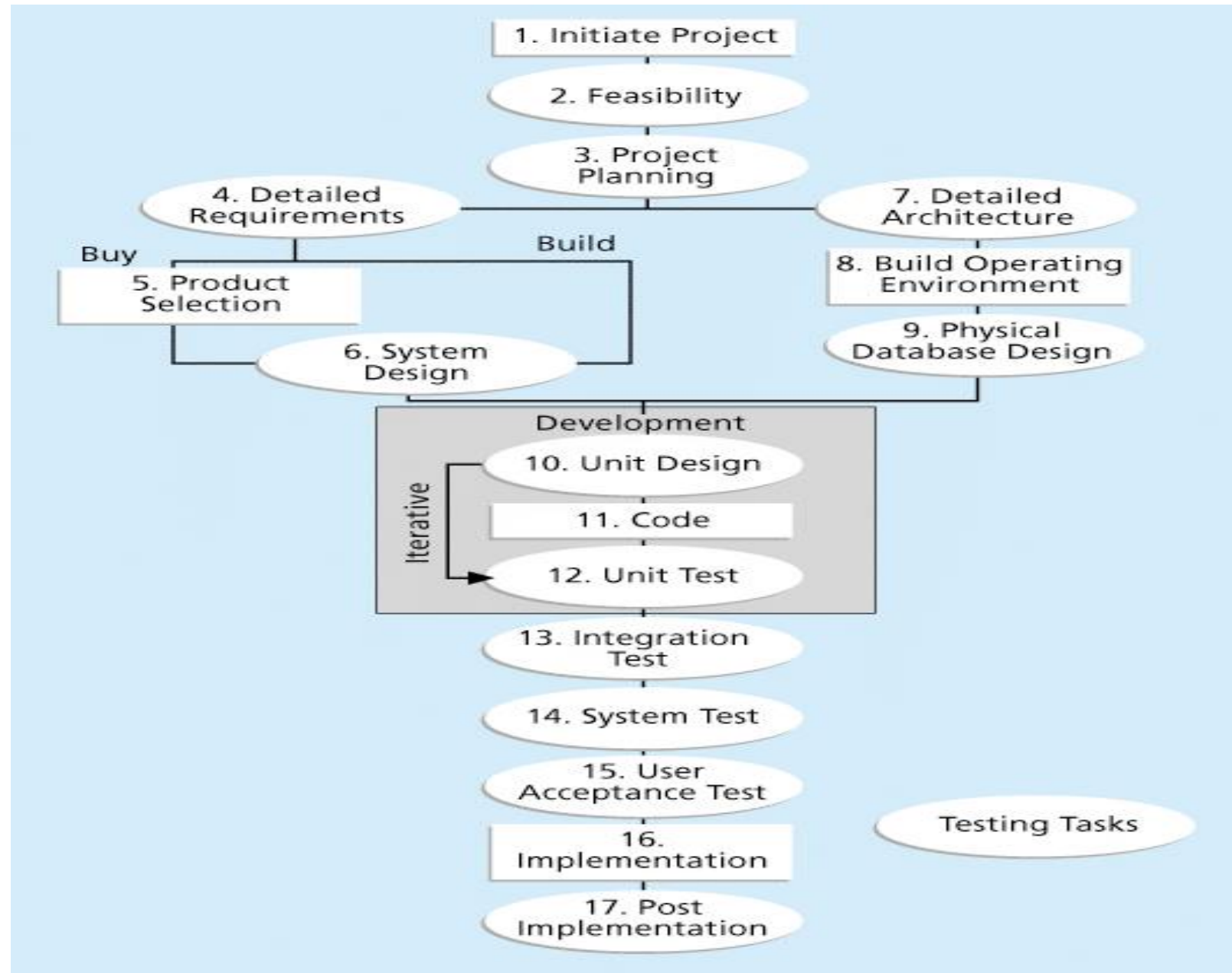
Statistical Sampling

- Statistical sampling involves choosing part of a population of interest for inspection.
- The size of a sample depends on how representative you want the sample to be.
- Be sure to consult with an expert when using statistical analysis

Testing

- Many IT professionals think of testing as a stage that comes near the end of IT product development.
- Testing should be done during almost every phase of the IT product development life cycle

Testing Tasks in the Software Development Life Cycle



ISO Standards

- ISO 9000 is a quality system standard that:
 - Is a three-part, continuous cycle of planning, controlling, and documenting quality in an organization.
 - Provides minimum requirements needed for an organization to meet its quality certification standards.
 - Helps organizations around the world reduce costs and improve customer satisfaction

Improving Information Technology Project Quality

- Several suggestions for improving quality for IT projects include:
 - Establish leadership that promotes quality
 - Understand the cost of quality
 - Focus on organizational influences and workplace factors that affect quality
 - Follow maturity models

Leadership

- As Joseph M. Juran said in 1945, “It is most important that top management be quality-minded. In the absence of sincere manifestation of interest at the top, little will happen below”*
- A large percentage of quality problems are associated with management, not technical issues.
 - *American Society for Quality (ASQ),
(www.asqc.org/about/history/juran.html).

The Cost of Quality

- The cost of quality is the cost of conformance plus the cost of nonconformance
 - Conformance means delivering products that meet requirements and fitness for use
 - Cost of nonconformance means taking responsibility for failures or not meeting quality expectations
- A study reported that software bugs cost the U.S. economy \$59.6 billion each year and that one third of the bugs could be eliminated by an improved testing infrastructure.

Five Cost Categories Related to Quality

- Prevention cost: Cost of planning and executing a project so it is error-free or within an acceptable error range.
- Appraisal cost: Cost of evaluating processes and their outputs to ensure quality. make user processors are inline, spend sufficient money in order to improve the processe
- Internal failure cost: Cost incurred to correct an identified defect before the customer receives the product.

Five Cost Categories Related to Quality

- External failure cost: Cost that relates to all errors not detected and corrected before delivery to the customer.
- Measurement and test equipment costs: Capital cost of equipment used to perform prevention and appraisal activities.

Organizational Influences, Workplace Factors, and Quality

- Study by DeMarco and Lister showed that organizational issues had a much greater influence on programmer productivity than the technical environment or programming languages.
- Programmer productivity varied by a factor of one to ten across organizations, but only by 21 percent within the same organization.
- Study found no correlation between productivity and programming language, years of experience, or salary.
- A dedicated workspace and a quiet work environment were key factors to improving programmer productivity

Chapter Summary

- Project quality management ensures that the project will satisfy the needs for which it was undertaken.
- Main processes include:
 - Plan quality
 - Perform quality assurance
 - Perform quality control



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Introduction to Research and Scholarly Communication



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What is a research

- The creation of new knowledge and/or the use of existing knowledge in a new and creative way to generate new concepts, methodologies and understandings.

What is a research paper

- A written document that presents the results of an original research or investigation on a particular topic.
- Content of a research paper:
 - Abstract - Provides a concise summary of the key points of the paper, including the research problem, objectives, methods, results, and conclusions.
 - Introduction - Provides an overview of the background of the topic, problem statement, significance of the research, research questions, and a preview of the remaining sections of the paper
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 - Methodology - Explains the research design and procedures
 - Results - Presents the findings of the study
 - Discussion - Interprets and contextualizes the results considering the research questions and existing knowledge
 - Conclusion - Summarizes the key findings, interpret their implications, and provide practical recommendations for future research or practice
 - Reference - Lists the sources cited in the research paper

Abstract Vs Conclusion

Aspect	Abstract	Conclusion
Length	Usually limited to 150-250 words, depending on the journal or conference requirements.	Can vary in length depending on the complexity of the study, but typically longer than the abstract.
Content	Provides a brief summary of the research problem, objectives, methods, results, and conclusions of the study.	Summarizes the key findings of the study and explains their implications, limitations, and recommendations for future research or practice.
Audience	Intended for readers who want to quickly understand the main points of the paper before reading the full text.	Intended for readers who have already read the full paper and want to understand the main implications and contributions of the study.

How to identify a research problem

- Involves finding a gap in existing knowledge or a question that has not yet been answered.
- Steps to follow:
 - Identify a general area of interest
 - Conduct a literature review and identify gaps in knowledge or areas that have not been explored in depth
 - Based on your review of the literature, brainstorm potential research questions that could be investigated to address the gaps or unanswered questions in the field.
 - Evaluate the feasibility of investigating each potential research question, considering factors such as available resources, data availability, and ethical considerations.
 - Refine the research questions based on feedback from peers or experts in the field, and ensure that they are specific, focused, and answerable
 - Consider the potential implications and impact of each research question and determine the possible consequences and benefits that they may have on various stakeholders.

Conference Papers Vs Journal Articles

Aspect	Conference Papers	Journal Articles
Publication Format	Proceedings of a conference (book or online)	Scholarly journals (printed or online)
Page Size	Range from 6 to 10 pages	Range from around 10 to 30 or more pages
Presentation Requirement	Presented at a conference	Not required to be presented
Peer-Review Process	Subject to a streamlined review process that can take a few weeks to a few months	Subject to a more rigorous review process that can take several months to a year or more

How to find Scopus indexed papers

- Go to Scopus by typing the following : <https://www.scopus.com/sources.uri>



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Create account

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Enter title

Find sources



Improved Citescore

We have updated the CiteScore methodology to ensure a more robust, stable and comprehensive metric which provides an indication of research impact, earlier. The updated methodology will be applied to the calculation of CiteScore, as well as retroactively for all previous CiteScore years (ie. 2018, 2017, 2016...). The previous CiteScore values have been removed and are no longer available.

[View CiteScore methodology.](#)



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2021



	Source title ↓	CiteScore ↓	Highest percentile ↓	Citations 2018-21 ↓	Documents 2018-21 ↓	% Cited ↓
☐ 1	Ca-A Cancer Journal for Clinicians	716.2	99% 1/360 Oncology	76,632	107	91

How to find Scopus indexed papers

- Select “*Subject area*” from the dropdown list
- Select a subject area that matches your research topic from the “*Enter subject area*” field

The screenshot shows the Scopus search interface. On the left, there is a 'Subject area' dropdown menu with a blue arrow icon. Below it, there are two informational boxes: one with a blue 'i' icon titled 'Improved Citescore' and another with a red '!' icon titled 'Invalid subject area'. The 'Invalid subject area' box shows the text 'Subject area software engineeri'. To the right of the dropdown menu is a search bar labeled 'Enter subject area'. Below the search bar is a list of subject areas, each with a checkbox. The list includes: Computer Science, Artificial Intelligence, Computational Theory and Mathematics, Computer Graphics and Computer-Aided Design, Computer Networks and Communications, Computer Science (miscellaneous), Computer Science Applications, Computer Vision and Pattern Recognition, General Computer Science, Hardware and Architecture, Human-Computer Interaction, and Information Systems. A vertical scrollbar is visible on the right side of the list. On the far right, there are two empty boxes with a blue border and a red border, each containing a close button (X).

Subject area

Enter subject area

Improved Citescore
We have updated the CiteScore of research impact, earlier. The previous CiteScore years (ie. 2019, 2020) are no longer available. View CiteScore methodology.

Invalid subject area
Subject area software engineeri

Computer Science

Artificial Intelligence

Computational Theory and Mathematics

Computer Graphics and Computer-Aided Design

Computer Networks and Communications

Computer Science (miscellaneous)

Computer Science Applications

Computer Vision and Pattern Recognition

General Computer Science

Hardware and Architecture

Human-Computer Interaction

Information Systems

Project Communication Management

Learning Objectives

- Understand the importance of good communications on projects and the need to develop soft skills, especially for IT project managers and their teams.
- Explain the elements of planning project communications and how to create a communications management plan.
- Describe how to manage communications, including communication technologies, media, and performance reporting.

Learning Objectives (cont'd)

- Discuss methods for controlling communications to ensure that information needs are met throughout the life of the project.
- List various methods for improving project communications, such as running effective meetings, using various technologies effectively, and using templates.

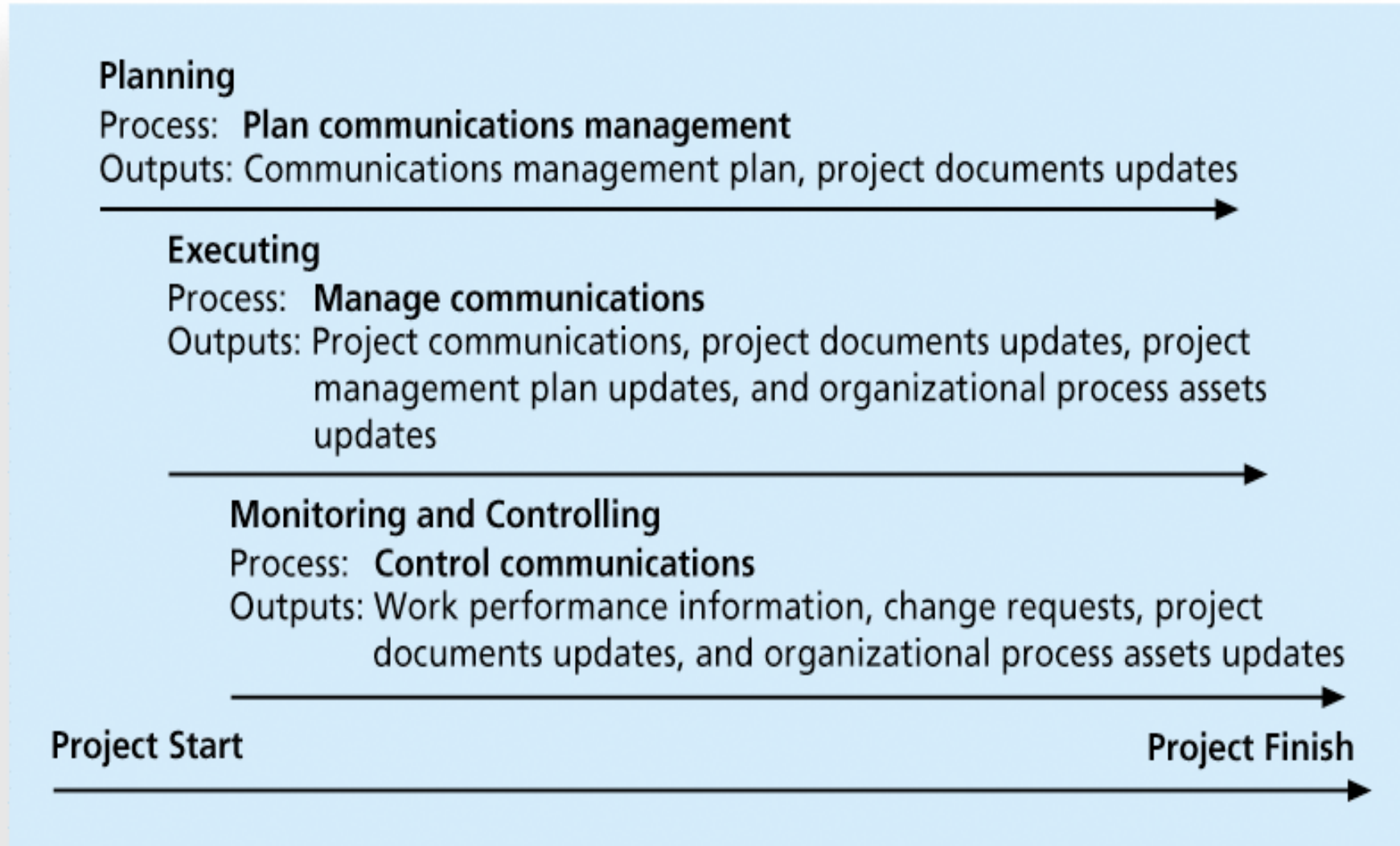
Importance of Good Communications

- The greatest threat to many projects is a failure to communicate.
- Our culture does not portray IT professionals as being good communicators.
- Research shows that IT professionals must be able to communicate effectively to succeed in their positions.
- Strong verbal and non-technical skills are a key factor in career advancement for IT professionals.

Project Communications Management Processes

- **Planning communications management:** Determining the information and communications needs of the stakeholders.
- **Managing communications:** Creating, distributing, storing, retrieving, and disposing of project communications based on the communications management plan.
- **Controlling communications:** Monitoring and controlling project communications to ensure that stakeholder communication needs are met.

Project Communications Management Summary



Importance of Face-to-Face Communication

- Research says that in a face-to-face interaction:
 - 58 percent of communication is through body language.
 - 35 percent of communication is through how the words are said.
 - 7 percent of communication is through the content or words that are spoken.
- Pay attention to more than just the actual words someone is saying.
- A person's tone of voice and body language say a lot about how he or she really feels.

Encouraging More Face-to-Face Interactions

- Short, frequent meetings are often very effective in IT projects.
- Stand-up meetings force people to focus on what they really need to communicate.
- Some companies have policies preventing the use of e-mail between certain hours or even entire days of the week.

Distributing Information in an Effective and Timely Manner

- Don't bury crucial information.
- Don't be afraid to report bad information.
- Oral communication via meetings and informal talks helps bring important information—good and bad—out into the open.

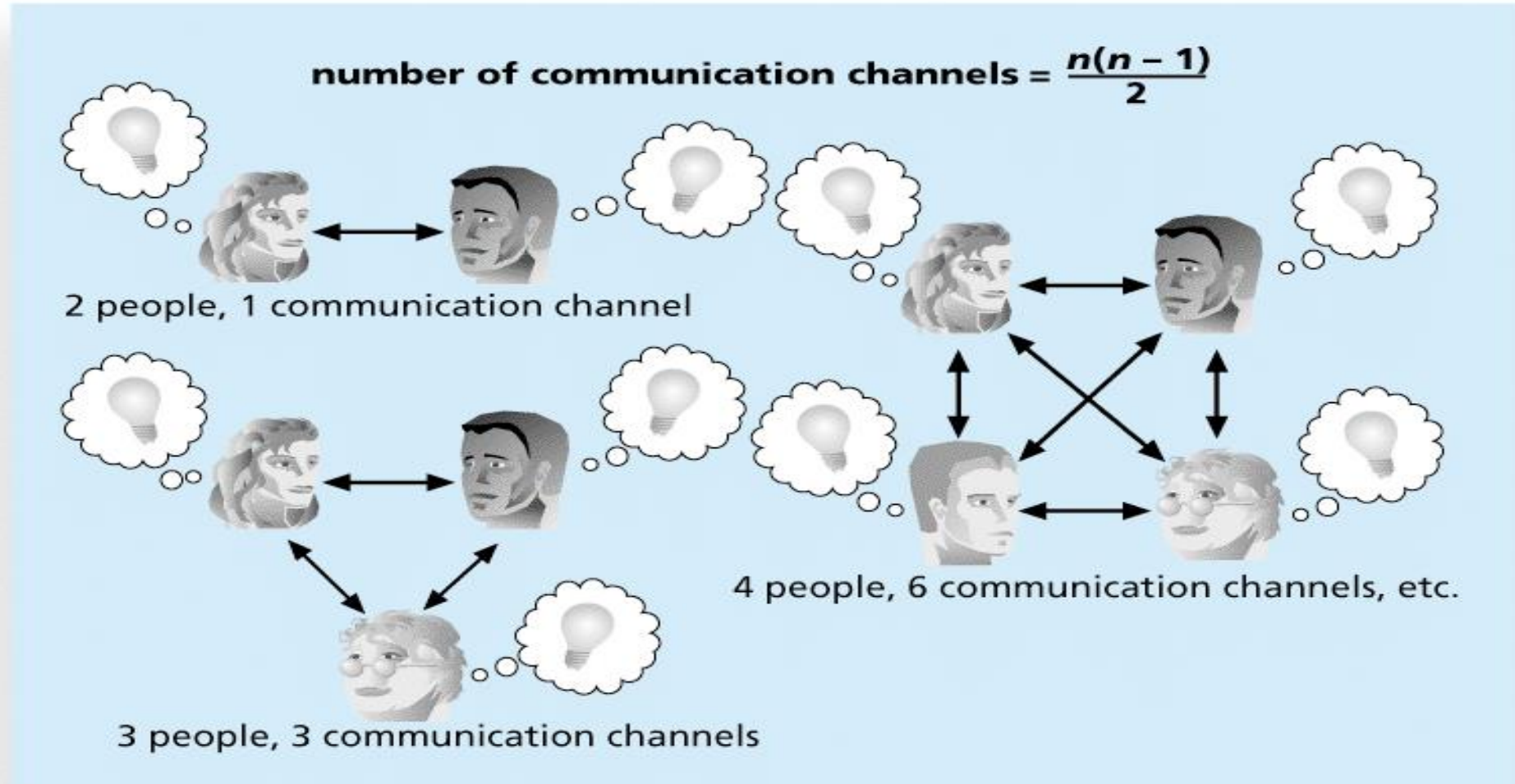
Other Communication Considerations

- Rarely does the receiver interpret a message exactly as the sender intended. But both the sender and the receiver are responsible for ensuring that the communication delivered the information intended.
- Geographic location and cultural background affect the complexity of project communications:
 - Different working hours
 - Language barriers
 - Different cultural norms

Determining the Number of Communications Channels

- As the number of people involved increases, the complexity of communications increases because there are more communications channels or pathways through which people can communicate.
- Number of communications channels = $n(n-1)/2$
where n is the number of people involved

The Impact of the Number of People on Communications Channels



Planning Communications Management

- Every project should include some type of **communications management** plan, a document that guides project communications.
- The communications management plan varies with the needs of the project, but some type of written plan should always be prepared.
- For small projects, the communications management plan can be part of the team contract.
- For large projects, it should be a separate document.

Communications Management Plan Contents

1. Stakeholder communications requirements
2. Information to be communicated, including format, content, and level of detail
3. Who will receive the information and who will produce it
4. Suggested methods or technologies for conveying the information
5. Frequency of communication
6. Escalation procedures for resolving issues
7. Revision procedures for updating the communications management plan
8. A glossary of common terminology

Sample Stakeholder Analysis for Project Communications

Stakeholders	Document Name	Document Format	Contact Person	Due
Customer management	Monthly status report	Hard copy and meeting	Tina Erndt, Tom Silva	First of month
Customer business staff	Monthly status report	Hard copy	Julie Grant, Sergey Cristobal	First of month
Customer technical staff	Monthly status report	E-mail	Li Chau, Nancy Michaels	First of month
Internal management	Monthly status report	Hard copy and meeting	Bob Thomson	First of month
Internal business and technical staff	Monthly status report	Intranet	Angie Liu	First of month
Training subcontractor	Training plan	Hard copy	Jonathan Kraus	November 1
Software subcontractor	Software implementation plan	E-mail	Najwa Gates	June 1

Comments: Put the titles and dates of documents in e-mail headings and have recipients acknowledge receipt.

Improving Team Collaboration and Communication

- Promote transparency and accountability
- Set clear expectations and goals
- Establish effective communication channels and protocols
- Provide regular feedback and recognition
- Foster a culture of trust, respect, and mutual support.
- Facilitate team building activities and promote opportunities for social interaction and bonding among team members.

Managing Communications

- Managing communications is a large part of a project manager's job.
- Getting project information to the right people at the right time and in a useful format is just as important as developing the information in the first place.
- Important considerations include the use of technology, the appropriate methods and media to use, and performance reporting.

Reporting Performance

Performance reporting keeps stakeholders informed about how resources are being used to achieve project objectives:

- **Status reports** describe where the project stands at a specific point in time.
- **Progress reports** describe what the project team has accomplished during a certain period of time.
- **Forecasts** predict future project status and progress based on past information and trends.

Controlling Communications

- The main goal of controlling communications is to ensure the optimal flow of information throughout the entire project life cycle.
- The project manager and project team should use their various reporting systems, expert judgment, and meetings to assess how well communications are working. If problems exist, the project manager and team need to take action, which often requires changes to the earlier processes of planning and managing project communications.
- It is often beneficial to have a facilitator from outside the project team assess how well communications are working.

Chapter Summary

- The goal of project communications management is to ensure timely and appropriate generation, collection, dissemination, storage, and disposition of project information.
- Main process include:
 - Plan communications management
 - Manage communications
 - Control communications



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Introduction to Research and Scholarly Communication

Dr. Dilshan De Silva



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Citing Vs Listing

Citing

Complexity metrics have a lot of potential uses which include: provision of feedback during a software project to help control the design activity, and provision of detailed information about software modules to help pinpoint areas of potential instability during testing and maintenance. Cyclomatic complexity is the most widely used complexity metric for computer software [12]. It is a software metric that provides a quantitative measure of the logical complexity of a program. The introduction of cognitive informatics to the software engineering domain through the work of Wang [13] has brought about the emergence of a new set of complexity metrics referred to as cognitive complexity metrics. These metrics introduce cognitive weights - which define the effort required, relative time or extent of difficulty in comprehending software. In cognitive informatics, the functional complexity of software in design and comprehension depends on three key elements namely: its input, internal processing and output [14]. Initially three basic control structures (BCS), branch, iteration and sequence were identified [15]. However, the work of Shao and Wang [14] modified these BCSs and introduced what obtains in Table 1. These BCSs are the fundamental logic building blocks of software.

Listing

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- [11] T. J. McCabe and A. H. Watson, "Software complexity," *Crosstalk*, vol. 7, no. 12, pp. 5-9, 1994.
- [12] R. S. Pressman, *Software Engineering: A Practitioner's Approach*. New York, NY, USA: McGraw-Hill, 2005, pp. 649-672.
- [13] Y. Wang, "On the cognitive informatics foundations of software engineering," in *Proc. ICCI*, 2004, pp. 22-31.
- [14] J. Shao and Y. Wang, "A new measure of software complexity based on cognitive weights," *Electr. Comput. Eng., Can. J.*, vol. 28, no. 2, pp. 69-74, Apr. 2003.
- [15] C. A. R. Hoare *et al.*, "Laws of programming," *Comm ACM*, vol. 30, no. 8, pp. 672-686, Aug. 1987.

Identifying a research problem

- Steps to follow:
 - Identify a general area of interest
 - Conduct a literature review and identify gaps in knowledge or areas that have not been explored in depth
 - Based on your review of the literature, brainstorm potential research questions that could be investigated to address the gaps or unanswered questions in the field.
 - Evaluate the feasibility of investigating each potential research question, considering factors such as available resources, data availability, and ethical considerations.
 - Refine the research questions based on feedback from peers or experts in the field, and ensure that they are specific, focused, and answerable
 - Consider the potential implications and impact of each research question and determine the possible consequences and benefits that they may have on various stakeholders.

Guidelines

- Do not write a research paper in the first person. Instead, use the third person.
- Do not use the speaking style in research paper writing.
- Do not have lengthy sentences.
- Do not use words such as "so". Instead, use words such as “thus”, “hence”, “therefore”, “accordingly”, etc.
- Be specific. Do not say the following figure, table, or sub-section. Instead, use numbers. For example, Table 1, Fig. 1, sub-section 1.2, etc.

Guidelines

- The text of figures and tables should be clearly visible when the document is viewed at 100%.
- Do not include the tables and equations as pictures.
- Number the equations in the paper.
- A reference should be given to figures, tables, and equations from the body of the paper before including them.
- It is possible to directly copy and paste sentences from other sections of the paper into the abstract

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Aspect	Conference Papers	Journal Articles
Publication Format	Proceedings of a conference (book or online)	Scholarly journals (printed or online)
Page Size	Range from 6 to 8 pages	Range from around 10 to 20 or more pages
Presentation Requirement	Presented at a conference	Not required to be presented
Peer-Review Process	Subject to a streamlined review process that can take a few weeks to a few months	Subject to a more rigorous review process that can take several months to a year or more

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- Select “*Subject area*” from the first dropdown list
- Select a subject area that matches your research topic from the “*Enter subject area*” field

Subject area Enter subject area ☐ Spectroscopy ☒ Computer Science ☐ Artificial Intelligence ☐ Computational Theory and Mathematics ☐ Computer Graphics and Computer-Aided Design ☐ Computer Networks and Communications ☐ Computer Science (miscellaneous) ☐ Computer Science Applications ☐ Computer Vision and Pattern Recognition ☐ General Computer Science Improved Citescore We have updated the CiteScore of research impact, earlier. The previous CiteScore years (ie. 20 View CiteScore methodology. refine list	Subject area Enter subject area ☐ Cratology ☒ Multidisciplinary ☐ Multidisciplinary ☒ Neuroscience ☐ Behavioral Neuroscience ☐ Biological Psychiatry ☐ Cellular and Molecular Neuroscience Improved Citescore We have updated the CiteScore of research impact, earlier. The previous CiteScore years (ie. 20 View CiteScore methodology.	Subject area Enter subject area ☐ Social Psychology ☒ Social Sciences ☐ Anthropology ☐ Archeology ☐ Communication ☐ Cultural Studies ☐ Demography ☐ Development ☐ Education ☐ Gender Studies ☐ General Social Sciences ☐ Geography, Planning and Development Improved Citescore We have updated the CiteScore of research impact, earlier. The previous CiteScore years (ie. 20 View CiteScore methodology. refine list Clear filters
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**Project
Risk
Management**



Learning Objectives

- Understand risk and the importance of good project risk management.
- Discuss the elements of planning risk management and the contents of a risk management plan.
- List common sources of risks on information technology (IT) projects.
- Describe the process of identifying risks and create a risk register.



Learning Objectives (cont'd)

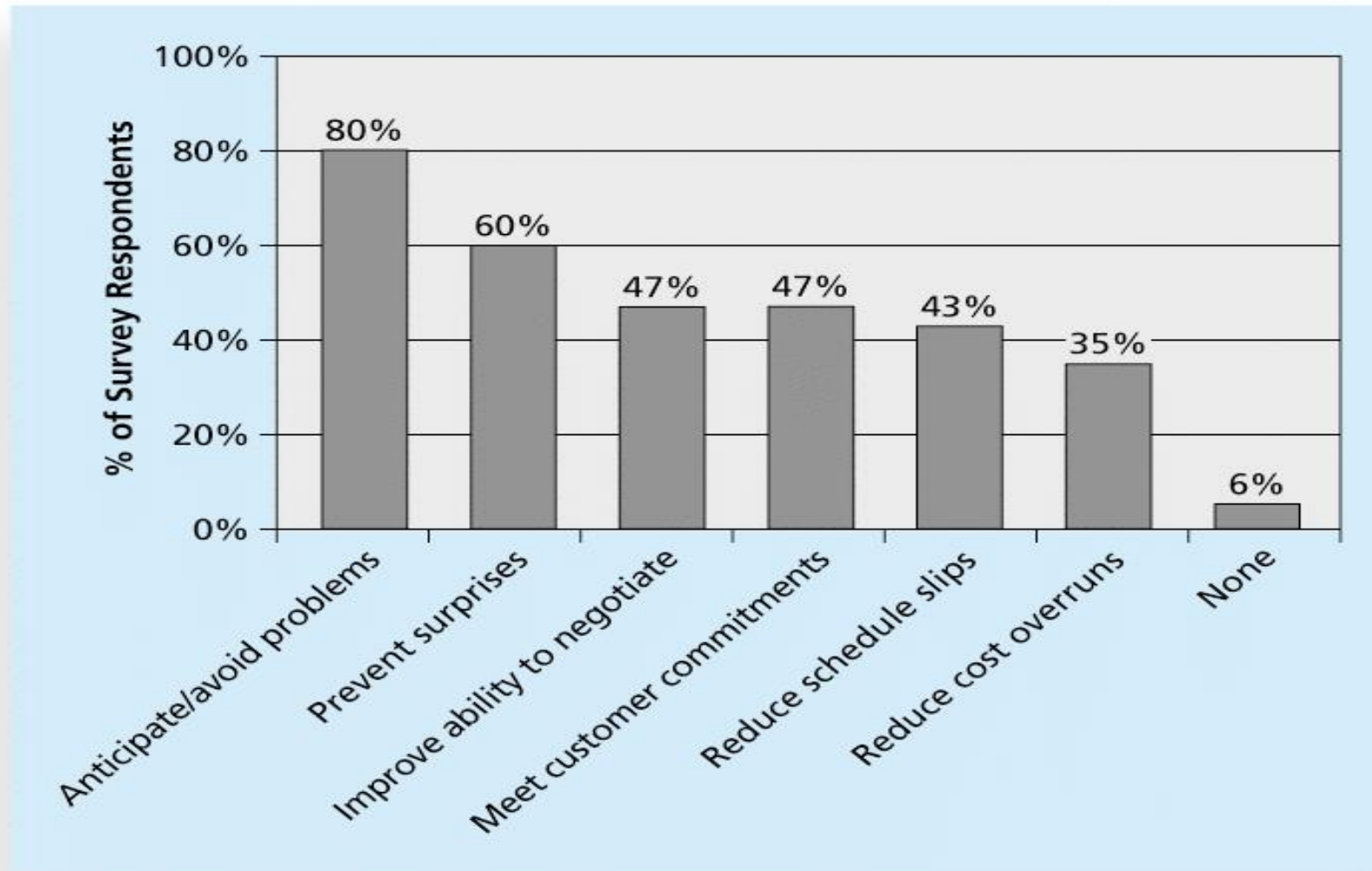
- Discuss qualitative risk analysis and explain how to calculate risk factors, create probability/impact matrixes, and apply the Top Ten Risk Item Tracking technique to rank risks.
- Explain quantitative risk analysis and how to apply decision trees, simulation, and sensitivity analysis to quantify risks.
- Provide examples of using different risk response planning strategies to address both negative and positive risks.
- Discuss how to control risks.



The Importance of Project Risk Management

- Project risk management is the art and science of identifying, analyzing, and responding to risk throughout the life of a project and in the best interests of meeting project objectives.
- Risk management is often overlooked in projects, but it can help improve project success by helping select good projects, determining project scope, and developing realistic estimates.

Benefits from Software Risk Management Practices*



*Source: Kulik and Weber, KLCI Research Group



Negative Risk

- A dictionary definition of risk is “the possibility of loss or injury”.
- Negative risk involves understanding potential problems that might occur in the project and how they might impede project success.
- Negative risk management is like a form of insurance; it is an investment.



Risk Can Be Positive

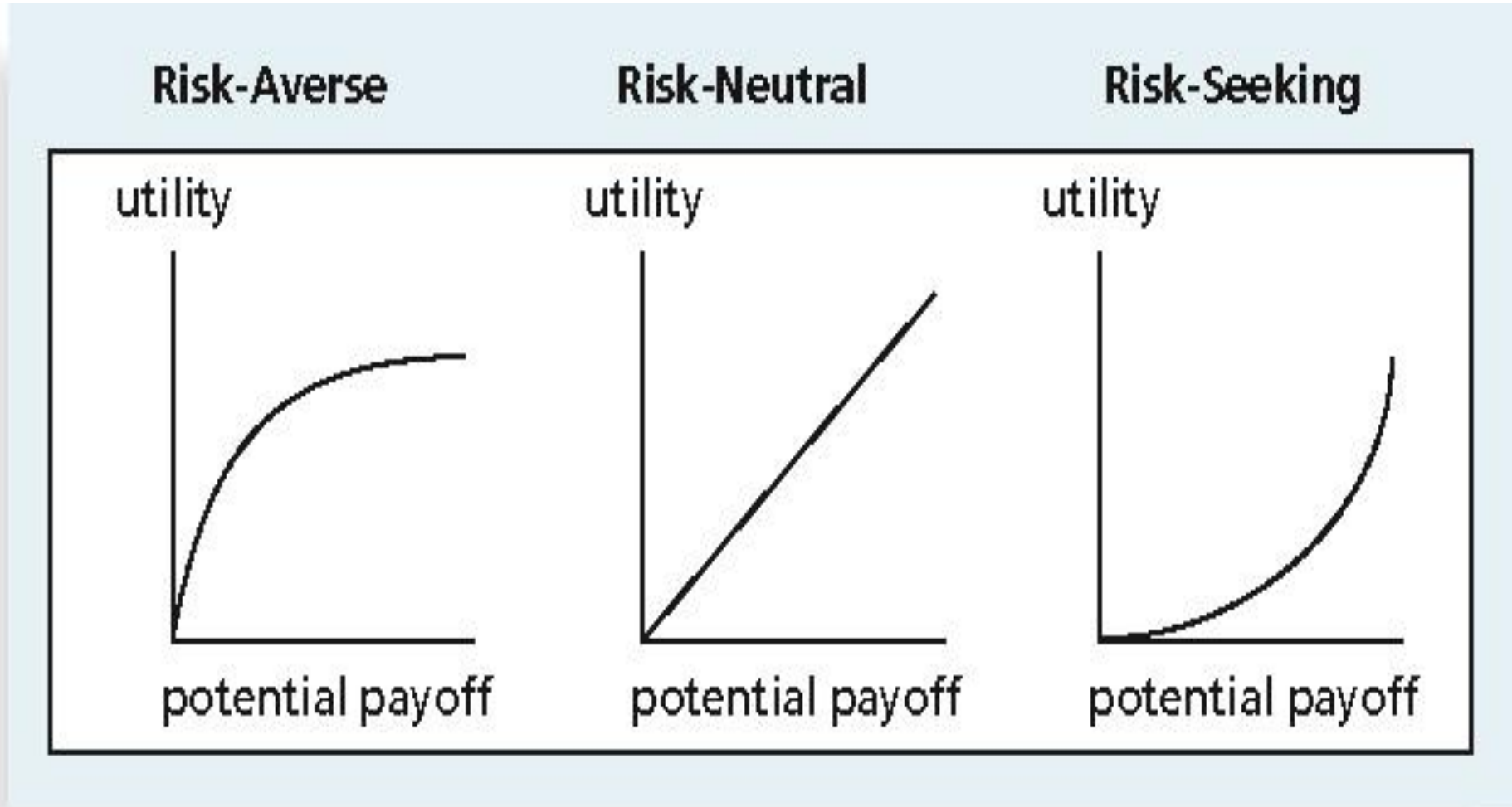
- Positive risks are risks that result in good things happening; sometimes called opportunities.
- A general definition of project risk is an uncertainty that can have a negative or positive effect on meeting project objectives.
- The goal of project risk management is to minimize potential negative risks while maximizing potential positive risks.



Risk Utility

- **Risk utility** or **risk tolerance** is the amount of satisfaction or pleasure received from a potential payoff
 - Utility rises at a decreasing rate for people who are risk-averse.
 - Those who are risk-seeking have a higher tolerance for risk and their satisfaction increases when more payoff is at stake.
 - The risk-neutral approach achieves a balance between risk and payoff.

Risk Utility Function and Risk Preference





Project Risk Management Processes

- **Planning risk management** : Deciding how to approach and plan the risk management activities for the project.
- **Identifying risks**: Determining which risks are likely to affect a project and documenting the characteristics of each.
- **Performing qualitative risk analysis**: Prioritizing risks based on their probability and impact of occurrence.



Project Risk Management Processes (cont'd)

- **Performing quantitative risk analysis:** Numerically estimating the effects of risks on project objectives.
- **Planning risk responses:** Taking steps to enhance opportunities and reduce threats to meeting project objectives.
- **Controlling risk:** Monitoring identified and residual risks, identifying new risks, carrying out risk response plans, and evaluating the effectiveness of risk strategies throughout the life of the project.

Project Risk Management Summary

Planning

Process: **Plan risk management**

Outputs: Risk management plan

Process: **Identify risks**

Outputs: Risk register

Process: **Perform qualitative risk analysis**

Outputs: Project documents updates

Process: **Perform quantitative risk analysis**

Outputs: Project documents updates

Process: **Plan risk responses**

Outputs: Project management plan updates, project documents updates

Monitoring and Controlling

Process: **Control risks**

Outputs: Work performance information, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish



Planning Risk Management

- The main output of this process is a **risk management plan**—a plan that documents the procedures for managing risk throughout a project.
- The project team should review project documents and understand the organization's and the sponsor's approaches to risk.
- The level of detail will vary with the needs of the project.



Topics Addressed in a Risk Management Plan

- Methodology
- Roles and responsibilities
- Budget and schedule
- Risk categories
- Risk probability and impact
- Revised stakeholders' tolerances
- Tracking
- Risk documentation



Contingency, Fallback Plans & Contingency Reserves

- **Contingency plans** are predefined actions that the project team will take if an identified risk event occurs.
- **Fallback plans** are developed for risks that have a high impact on meeting project objectives and are put into effect if attempts to reduce the risk are not effective.
- **Contingency reserves** or **allowances** are provisions held by the project sponsor or organization to reduce the risk of cost or schedule overruns to an acceptable level; **management reserves** are funds held for unknown risks.

IT Success Potential Scoring Sheet

Success Criterion	Relative Importance
User Involvement	19
Executive Management support	16
Clear Statement of Requirements	15
Proper Planning	11
Realistic Expectations	10
Smaller Project Milestones	9
Competent Staff	8
Ownership	6
Clear Visions and Objectives	3
Hard-Working, Focused Staff	3
Total	100



Broad Categories of Risk

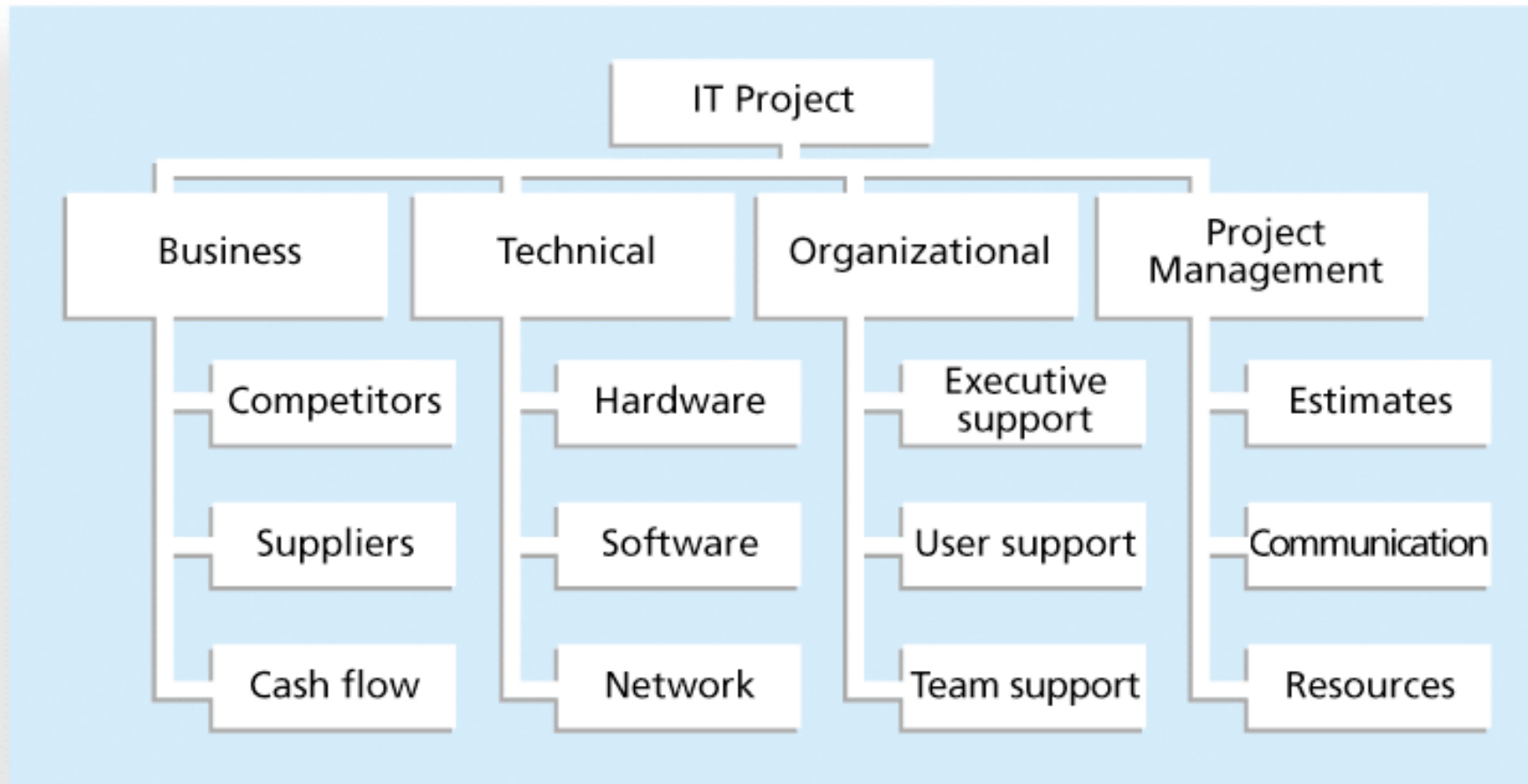
- Market risk
- Financial risk
- Technology risk
- People risk
- Structure/process risk



Risk Breakdown Structure

- A **risk breakdown structure** is a hierarchy of potential risk categories for a project.
- Similar to a work breakdown structure but used to identify and categorize risks.

Sample Risk Breakdown Structure





Identifying Risks

- Identifying risks is the process of understanding what potential events might hurt or enhance a particular project.
- Another consideration is the likelihood of advanced discovery.
- Risk identification tools and techniques include:
 - Brainstorming
 - The Delphi Technique
 - Interviewing
 - SWOT analysis



Brainstorming

- **Brainstorming** is a technique by which a group attempts to generate ideas or find a solution for a specific problem by amassing ideas spontaneously and without judgment.
- An experienced facilitator should run the brainstorming session.
- Be careful not to overuse or misuse brainstorming:
 - Psychology literature shows that individuals produce a greater number of ideas working alone than they do through brainstorming in small, face-to-face groups.
 - Group effects often inhibit idea generation.



Delphi Technique

- The **Delphi Technique** is used to derive a consensus among a panel of experts who make predictions about future developments.
- Provides independent and anonymous input regarding future events.
- Uses repeated rounds of questioning and written responses and avoids the biasing effects possible in oral methods, such as brainstorming.



Interviewing

- **Interviewing** is a fact-finding technique for collecting information in face-to-face, phone, e-mail, or instant-messaging discussions.
- Interviewing people with similar project experience is an important tool for identifying potential risks.



SWOT Analysis

- SWOT analysis (strengths, weaknesses, opportunities, and threats) can also be used during risk identification.
- Helps identify the broad negative and positive risks that apply to a project.



Risk Register

- The main output of the risk identification process is a list of identified risks and other information needed to begin creating a risk register.
- **A risk register** is:
 - A document that contains the results of various risk management processes and that is often displayed in a table or spreadsheet format.
 - A tool for documenting potential risk events and related information.
- **Risk events** refer to specific, uncertain events that may occur to the detriment or enhancement of the project.



Risk Register Contents

- An identification number for each risk event
- A rank for each risk event
- The name of each risk event
- A description of each risk event
- The category under which each risk event falls
- The root cause of each risk



Risk Register Contents (cont'd)

- Triggers for each risk; **triggers** are indicators or symptoms of actual risk events.
- Potential responses to each risk.
- The **risk owner** or person who will own or take responsibility for each risk.
- The probability and impact of each risk occurring.
- The status of each risk.



Performing Qualitative Risk Analysis

- Assess the likelihood and impact of identified risks to determine their magnitude and priority.
- Risk quantification tools and techniques include:
 - Probability/impact matrixes
 - The Top Ten Risk Item Tracking
 - Expert judgment



Probability/Impact Matrix

- A **probability/impact matrix** or **chart** lists the relative probability of a risk occurring on one side of a matrix or axis on a chart and the relative impact of the risk occurring on the other.
- List the risks and then label each one as high, medium, or low in terms of its probability of occurrence and its impact if it did occur.
- Can also calculate **risk factors**:
 - Numbers that represent the overall risk of specific events based on their probability of occurring and the consequences to the project if they do occur.

Sample Probability/Impact Matrix

Probability	High	risk 6	risk 9	risk 1 risk 4
	Medium	risk 3 risk 7	risk 2 risk 5 risk 11	
	Low		risk 8 risk 10	risk 12
		Low	Medium	High
		Impact		



Performing Quantitative Risk Analysis

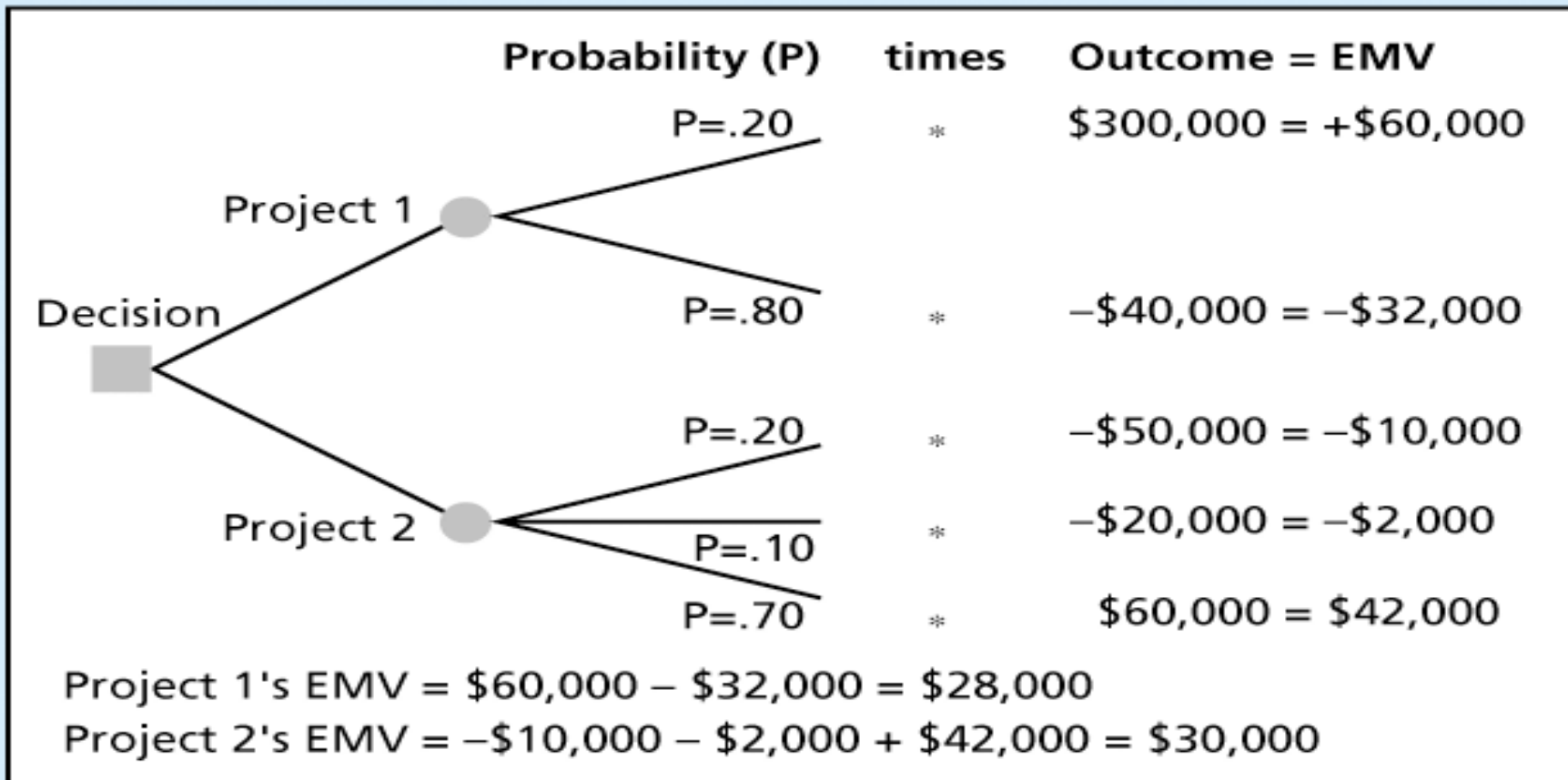
- Often follows qualitative risk analysis, but both can be done together.
- Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis.
- Main techniques include:
 - Decision tree analysis
 - Simulation
 - Sensitivity analysis



Decision Trees and Expected Monetary Value (EMV)

- A **decision tree** is a diagramming analysis technique used to help select the best course of action in situations in which future outcomes are uncertain.
- **Estimated monetary value (EMV)** is the product of a risk event probability and the risk event's monetary value.
- You can draw a decision tree to help find the EMV .

Expected Monetary Value (EMV) Example





Planning Risk Responses

- After identifying and quantifying risks, you must decide how to respond to them.
- Four main response strategies for negative risks:
 - Risk avoidance
 - Risk acceptance
 - Risk transference
 - Risk mitigation



Response Strategies for Positive Risks

- Risk exploitation
- Risk sharing
- Risk enhancement
- Risk acceptance



Residual and Secondary Risks

- It's also important to identify residual and secondary risks.
- **Residual risks** are risks that remain after all of the response strategies have been implemented.
- **Secondary risks** are a direct result of implementing a risk response.



Controlling Risks

- Involves executing the risk management process to respond to risk events and ensuring that risk awareness is an ongoing activity performed by the entire project team throughout the entire project.
- **Workarounds** are unplanned responses to risk events that must be done when there are no contingency plans.
- Main outputs of risk control are:
 - Work performance information
 - Change requests
 - Updates to the project management plan, other project documents, and organizational process assets

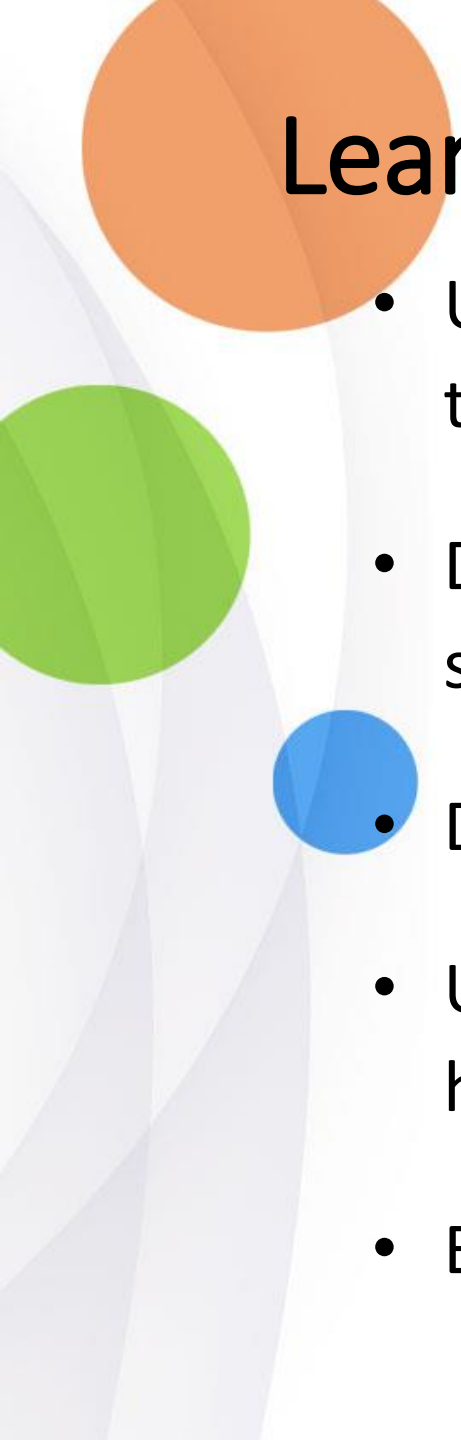


Chapter Summary

- Project risk management is the art and science of identifying, analyzing, and responding to risk throughout the life of a project and in the best interests of meeting project objectives.
- Main processes include:
 - Plan risk management
 - Identify risks
 - Perform qualitative risk analysis
 - Perform quantitative risk analysis
 - Plan risk responses
 - Control risks



Project Stakeholder Management



Learning Objectives

- Understand the importance of project stakeholder management throughout the life of a project.
- Discuss the process of identifying stakeholders, how to create a stakeholder register, and how to perform a stakeholder analysis.
- Describe the contents of a stakeholder management plan.
- Understand the process of managing stakeholder engagement and how to use an issue log effectively.
- Explain methods for controlling stakeholder engagement.



Project Stakeholder Management Processes

- **Identifying stakeholders:** Identifying everyone involved in the project or affected by it and determining the best ways to manage relationships with them.
- **Planning stakeholder management:** Determining strategies to effectively engage stakeholders.
- **Managing stakeholder engagement:** Communicating and working with project stakeholders to satisfy their needs and expectations, resolving issues , and fostering engagement in project decisions and activities.
- **Controlling stakeholder engagement:** Monitoring stakeholder relationships and adjusting plans and strategies for engaging stakeholders as needed.



Project Stakeholder Management Summary

Initiating

Process: **Identify stakeholders**

Outputs: Stakeholder register

Planning

Process: **Plan stakeholder management**

Outputs: Stakeholder management plan, project documents updates

Executing

Process: **Manage stakeholder engagement**

Outputs: Issue log, change requests, project management plan updates, project documents updates, organizational process assets updates

Monitoring and Controlling

Process: **Control stakeholder engagement**

Outputs: Work performance information, change requests, project documents updates, organizational process assets updates

Project Start

Project Finish



Identifying Stakeholders

- ***Internal project stakeholders*** generally include the project sponsor, project team, support staff, and internal customers for the project. Other internal stakeholders include top management, other functional managers, and other project managers because organizations have limited resources.
- ***External project stakeholders*** include the project's customers (if they are external to the organization), competitors, suppliers, and other external groups that are potentially involved in the project or affected by it, such as government officials and concerned citizens.



Stakeholder Register

- A stakeholder register includes basic information on stakeholders:
 - **Identification information:** The stakeholders' names, positions, locations, roles in the project, and contact information.
 - **Assessment information:** The stakeholders' major requirements and expectations, potential influences, and phases of the project in which stakeholders have the most interest.
 - **Stakeholder classification:** Is the stakeholder internal or external to the organization? Is the stakeholder a supporter of the project or resistant to it?

Sample Stakeholder Register

Name	Position	Internal/ External	Project Role	Contact Information
Stephen	VP of Operations	Internal	Project sponsor	stephen@globaloil.com
Betsy	CFO	Internal	Senior manager, approves funds	betsy@globaloil.com
Chien	CIO	Internal	Senior manager, PM's boss	chien@globaloil.com
Ryan	IT analyst	Internal	Team member	ryan@globaloil.com
Lori	Director, Accounting	Internal	Senior manager	lori@globaloil.com
Sanjay	Director, Refineries	Internal	Senior manager of largest refinery	sanjay@globaloil.com
Debra	Consultant	External	Project manager	debra@gmail.com
Suppliers	Suppliers	External	Supply software	suppliers@gmail.com



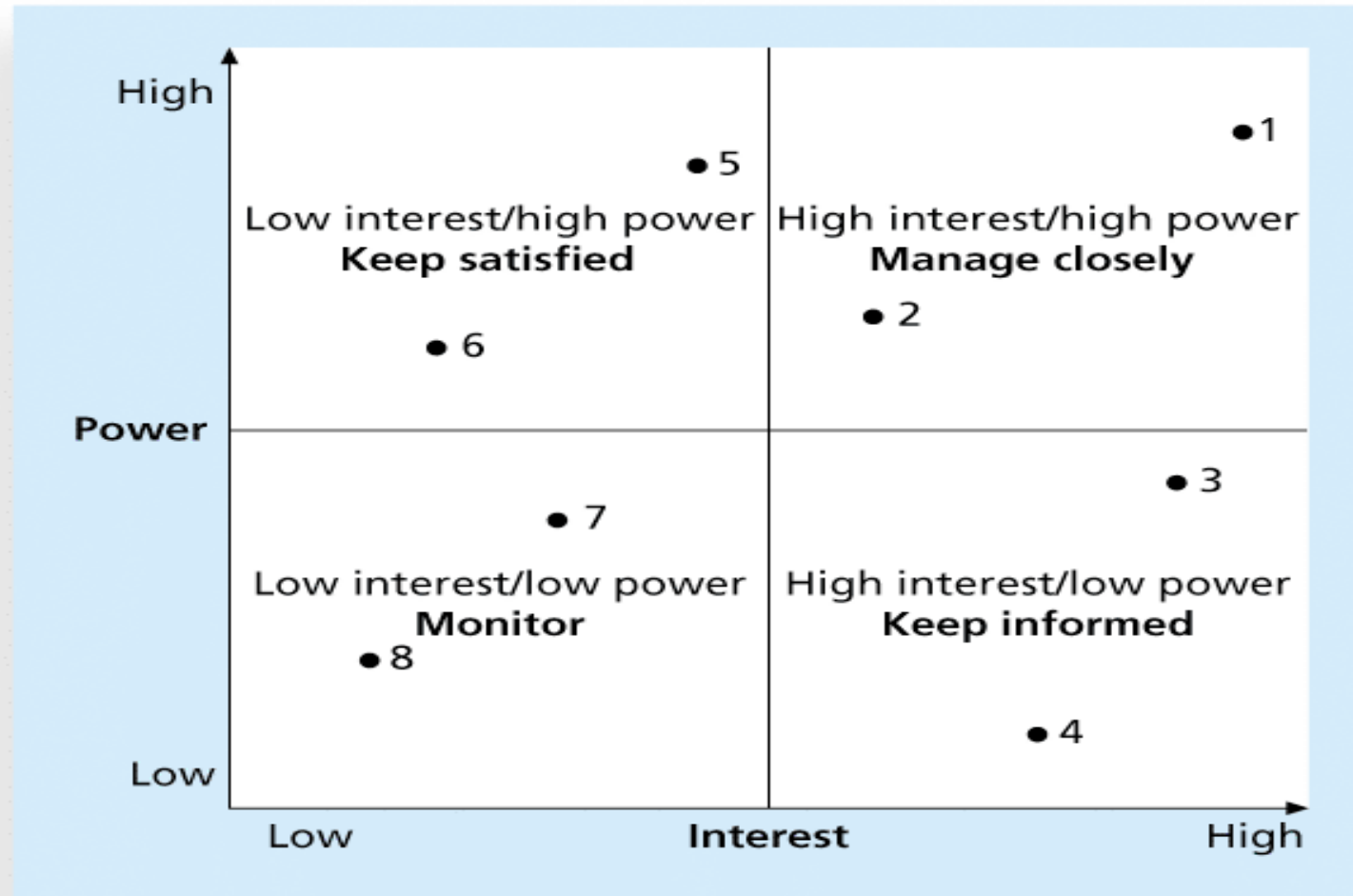
Classifying Stakeholders

- After identifying key project stakeholders, you can use different classification models to determine an approach for managing stakeholder relationships.
- A **power/interest grid** can be used to group stakeholders based on their level of authority (power) and their level of concern (interest) for project outcomes. It typically consists of four quadrants. Stakeholders in each quadrant may require different levels of attention and management approaches to effectively address their needs and concerns.

Power/Interest Grid

- In a software project, stakeholders can vary in terms of their power and interest. However, it is common for stakeholders in software projects to fall into the High Power/High Interest quadrant and the High Power/Low Interest quadrant of the Power/Interest grid.
- The High Power/High Interest quadrant typically includes stakeholders who have significant influence over the project and are highly engaged in its success. This may include project sponsors, senior management, project managers, or key users. They have the power to make important decisions and their active involvement is crucial for project success.
- The High Power/Low Interest quadrant may include stakeholders who have a high level of authority or power but have relatively lower interest or involvement in the day-to-day details of the project. These stakeholders may be senior executives or governing bodies who have the authority to allocate resources or approve major project milestones.
- However, it's important to note that stakeholder categorization can vary depending on the specific context and nature of the software project.

Power/Interest Grid





Planning Stakeholder Management

- After identifying and analyzing stakeholders, project teams should develop a plan for management them.
- The stakeholder management plan can include:
 - Current and desired engagement levels
 - Interrelationships between stakeholders
 - Communication requirements
 - Potential management strategies for each stakeholders
 - Methods for updating the stakeholder management plan

Sample Stakeholder Analysis

Name	Power/ Interest	Current Engagement	Potential Management Strategies
Stephen	High/high	Leading	Stephen can seem intimidating due to his physical stature and deep voice, but he has a great personality and sense of humor. He previously led a similar refinery upgrade program at another company and knows what he wants. Manage closely and ask for his advice as needed. He likes short, frequent updates in person.
Chien	High/ medium	Resistant	Chien is a very organized yet hardheaded man. He has been pushing corporate IT standards, and the system the PM and sponsor (Debra and Stephen) like best goes against those standards, even though it's the best solution for this project and the company as a whole. Need to convince him that this is okay and that people still respect his work and position.
Ryan	Medium/ high	Supportive	Ryan has been with the company for several years and is well respected, but he feels threatened by Debra. He also resents her getting paid more than he does. He wants to please his boss, Chien, first and foremost. Need to convince him that the suggested solution is in everyone's best interest.
Betsy	High/low	Neutral	Very professional, logical person. Gets along well with Chien. She has supported Debra in approving past projects with strong business cases. Provide detailed financial justification for the suggested solution to keep her satisfied. Also ask her to talk to Chien on Debra's behalf.



Managing Stakeholder Engagement

- Project success is often measured in terms of customer/sponsor satisfaction.
- Project sponsors often rank scope, time, and cost goals in order of importance and provide guidelines on how to balance the triple constraint.
- This ranking can be shown in an expectations management matrix to help clarify expectations.



The engagement level of the stakeholders

The engagement level of the stakeholders can be classified as follows:

- Unaware: Unaware of project and potential impacts.
- Resistant: Aware of project and potential impacts and resistant to change.
- Neutral: Aware of project yet neither supportive nor resistant.
- Supportive: Aware of project and potential impacts and supportive to change.
- Leading: Aware of project and potential impacts and actively engaged in ensuring the project is a success.

Expectations Management Matrix

Measure of Success	Priority	Expectations	Guidelines
Scope	1	The scope statement clearly defines mandatory requirements and optional requirements.	Focus on meeting mandatory requirements before considering optional ones. In this case, following corporate IT standards is optional.
Time	1	There is little give in the project completion date. The schedule is very realistic.	The project sponsor must be alerted if any issues might affect meeting schedule goals.
Cost	3	This project is crucial to the organization. If you can clearly justify the need for more funds, they can be made available.	There are strict rules for project expenditures and escalation procedures. Cost is very important, but it takes a back seat to meeting schedule and then scope goals.
Technology/standards	2	There are several potential solutions available, but only one that meets all of the sponsor's technical requirements, especially for accounting.	While corporate IT standards are important, an exception makes sense in this case.



Issue Logs

- Understanding the stakeholders' expectations can help in managing issues.
- Issues should be documented in an **issue log**, a tool used to document, monitor, and track issues that need resolution.
- Unresolved issues can be a major source of conflict and result in stakeholder expectations not being met.
- Issue logs can address other knowledge areas as well.



Controlling Stakeholder Engagement

- You cannot control stakeholders, but you can control their level of engagement.
- Engagement involves a dialogue in which people seek understanding and solutions to issues of mutual concern.
- It is important to set the proper tone at the start of a project.



Project Stakeholder Management Overview

Identify Stakeholders

Inputs

1. Project Charter
2. Procurement Documents
3. Enterprise Environment Factors
4. Organizational Process Assets

Tools and Techniques

1. Stakeholder analysis
2. Expert Judgement
3. Meetings

Outputs

1. Stakeholder Register

Plan Stakeholder Management

Inputs

1. Project Management Plan
2. Stakeholder Register
3. Enterprise Environment Factors
4. Organizational Process Assets

Tools and Techniques

1. Expert Judgement
2. Meetings
3. Analytical Techniques

Outputs

1. Stakeholder Management Plan

Manage Stakeholder Engagement

Inputs

1. Stakeholder Management Plan
2. Communication Management Plan
3. Change log
4. Organizational Process Assets

Tools and Techniques

1. Communication methods
2. Interpersonal Skills
3. Management Skills

Outputs

1. Issue Log
2. Change Request
3. Project Management Plan Updates
4. Project Document Updates
5. Organizational Project assets updates

Control Stakeholder Engagement

Inputs

1. Project Management Plan
2. Issue Log
3. Work Performance data
4. Project Document

Tools and Techniques

1. Information Management Systems
2. Expert Judgement
3. Meetings

Outputs

1. Work Performance Information
2. Change Requests
3. Project Management Plan Updates
4. Project Document Updates
5. Organizational Project assets updates



Chapter Summary

- Managing stakeholders is now the tenth knowledge management area.
- Processes include:
 - Identify stakeholders
 - Plan stakeholder management
 - Manage stakeholder engagement
 - Control stakeholder engagement